Set 2B: Op-Amp Circuits by Nodal Analysis

ECEN 325 — Lecture 2 (Operational Amplifiers) practice set

What this set trains. Op-amp circuits that are not one of the standard blocks. The recipe never changes: apply the virtual short, write KCL at every node that is not an op-amp output, and eliminate the unknown node voltages one at a time until only V_o and V_i are left. All op-amps are ideal. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-2 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1 tools for impedances, dividers, Bode plots and sinusoids):

- **T1** *Ideal op-amp with negative feedback:* no current into + or −; $V_- = V_+$ (virtual short). Its output is an ideal voltage source (loads on it do not matter).
- **T2** *Block formulas:* inverting $\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_{in}}$; non-inverting $\frac{V_o}{V_+} = 1 + \frac{Z_f}{Z_g}$ (with V_+ from the divider at the + input); a follower has gain 1. Several blocks in cascade: multiply.
- **T3** *KCL with impedances:* at a node, $\sum_k \frac{V_{node} - V_k}{Z_k} = 0$ (for a capacitor $\frac{1}{Z} = sC$). Write it at every node except op-amp outputs and source nodes.
- **T4** *Solving KCL equations:* pick the equation with the fewest unknowns, solve it for one node voltage, substitute into the rest; repeat until only V_o and V_i remain.
- **T5** *Finite gain:* $V_{out} = A(s)(V_+ - V_-)$, $A(s) = \frac{A_0}{1+s/\omega_b}$, $\omega_t = A_0\omega_b$. With $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$: $\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$.
- **T6** *Slew rate:* for $V_{out} = \hat{V}\sin\omega t$, the fastest change is $\left|\frac{dV_{out}}{dt}\right|_{max} = \omega\hat{V}$; no distortion requires $\omega\hat{V} \le SR$.

Steps:

1. *Sources.* DC and AC both present ⇒ superposition.
2. *DC circuit:* $V_i \to 0$, $C \to$ open; pass the DC through each block with its gain at $s = 0$.
3. *Op-amp rules:* confirm negative feedback; apply T1 (or T5 for a non-ideal op-amp).
4. *Transfer function:* standard blocks → T2; anything else → T3 + T4.
5. *Standard form*, 6. *Bode*, 7. *Output* — exactly as in Lecture 1.

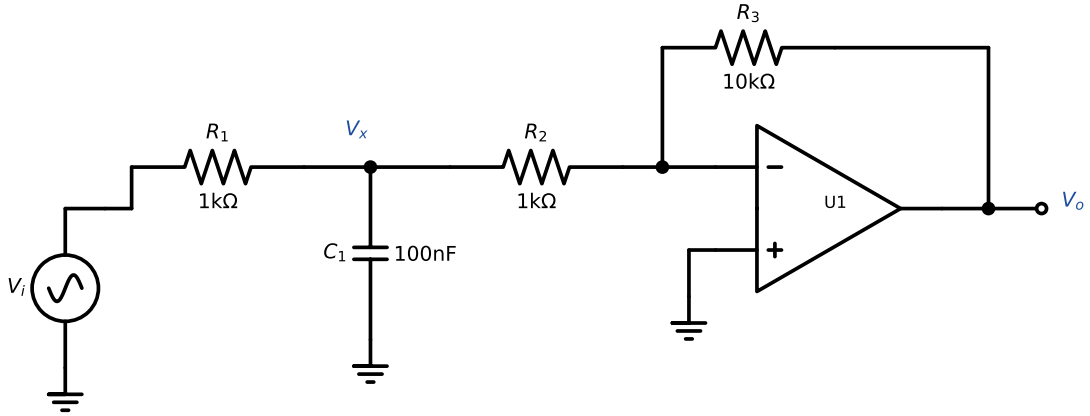
Variations: a design problem runs these steps backwards (Bode plot → $H(s)$ → blocks → component values); a slew-rate problem stops after Step 4 for every op-amp output and applies T6.

Problems

Easy problems

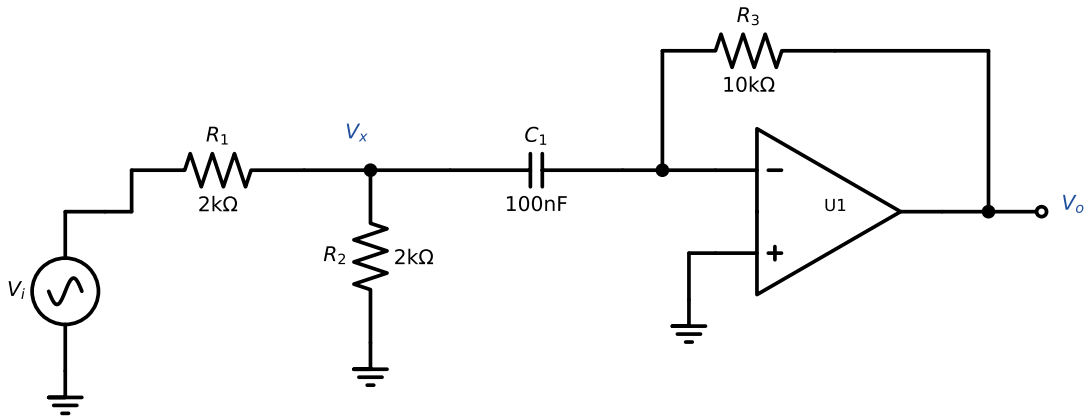
Problem 2B.1 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$.



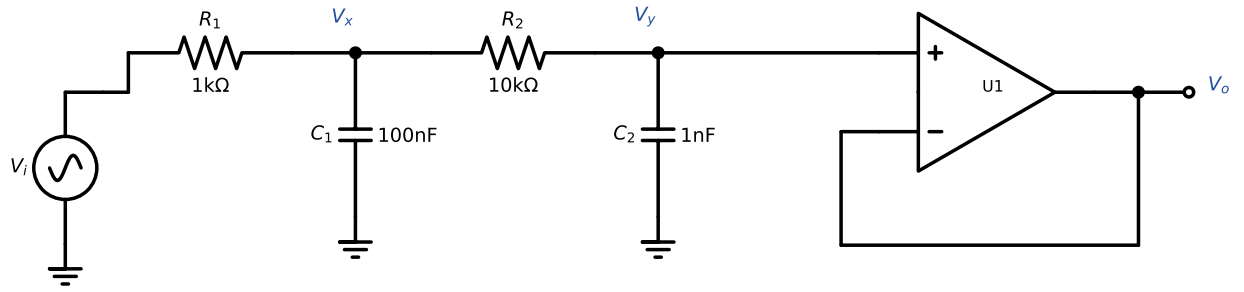
Problem 2B.2 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 500\,t)$.



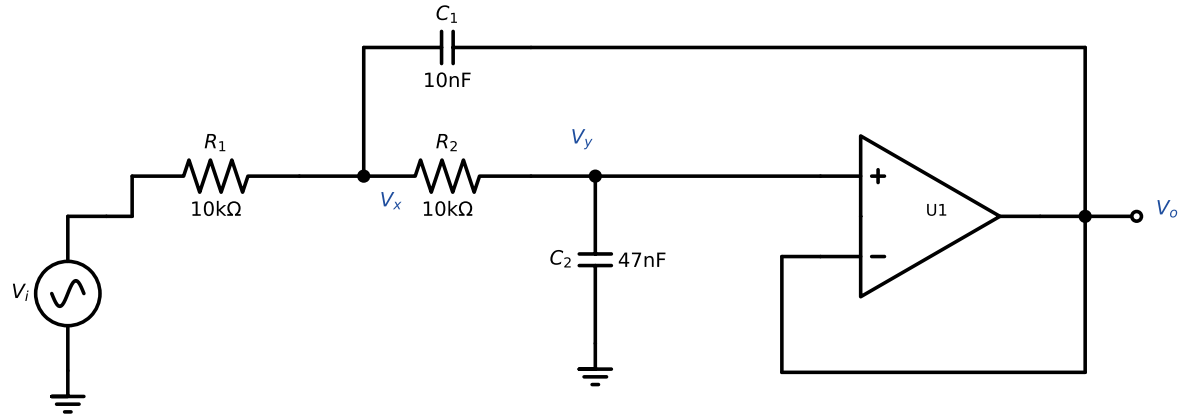
Problem 2B.3 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 5000\,t)$.



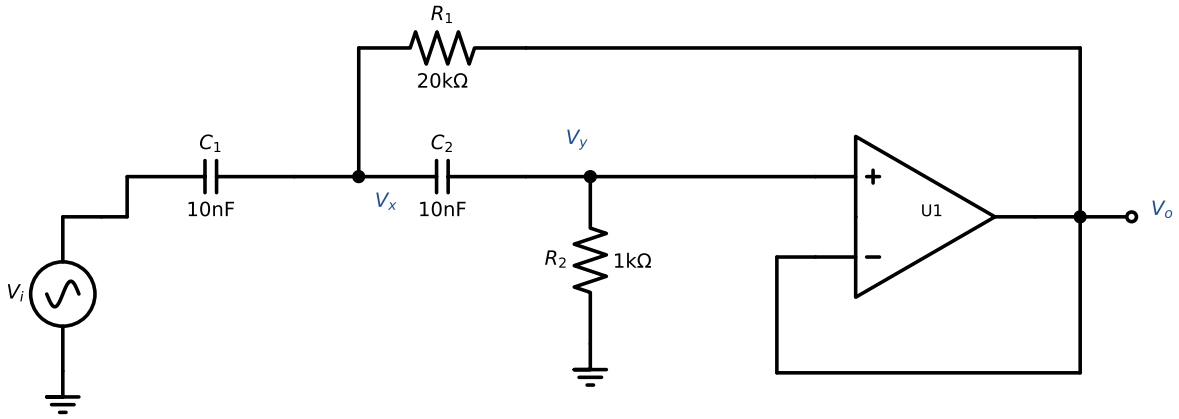
Problem 2B.4 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$.



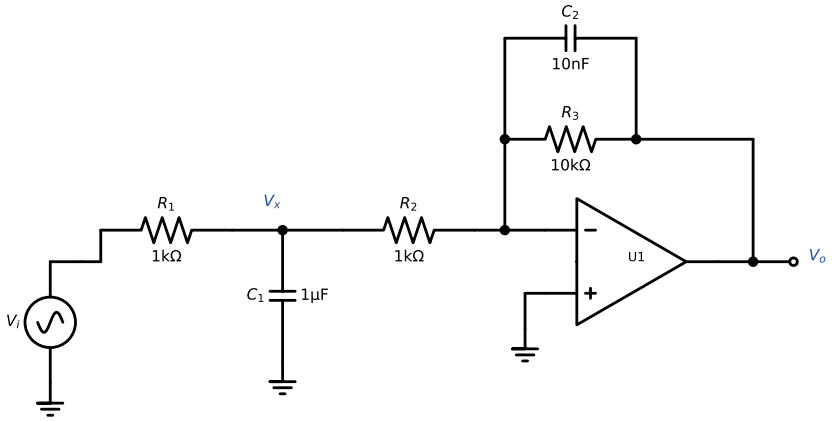
Problem 2B.5 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 50000\,t)$.



Problem 2B.6 — Easy

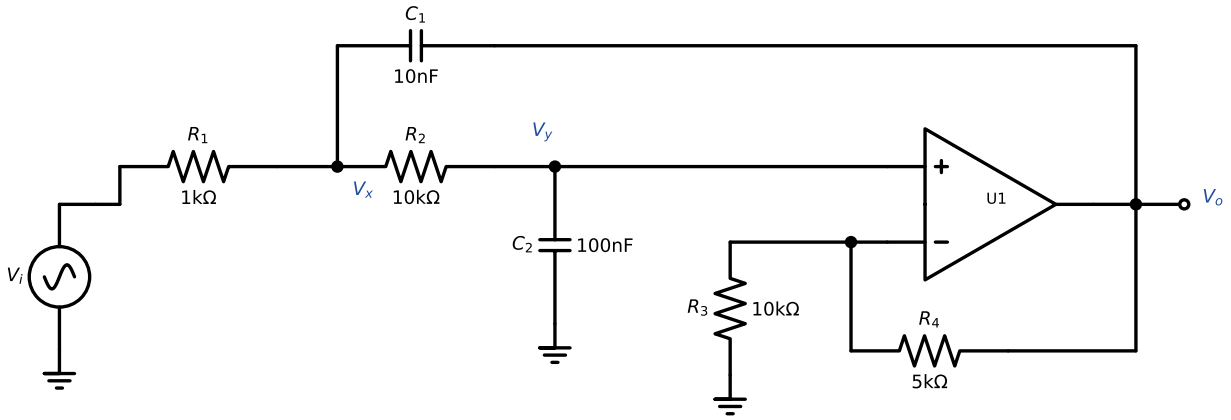
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\cos(2\pi \cdot 500\,t)$.



Medium problems

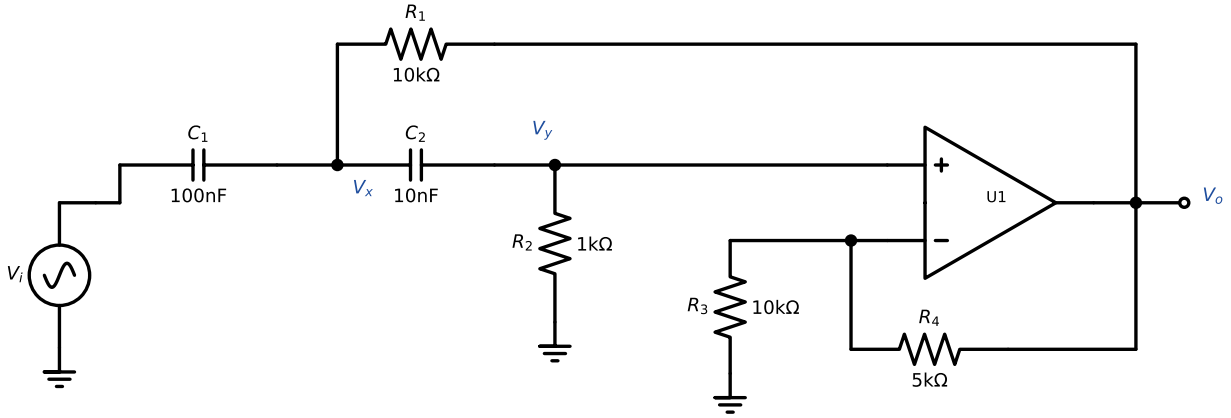
Problem 2B.7 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$.



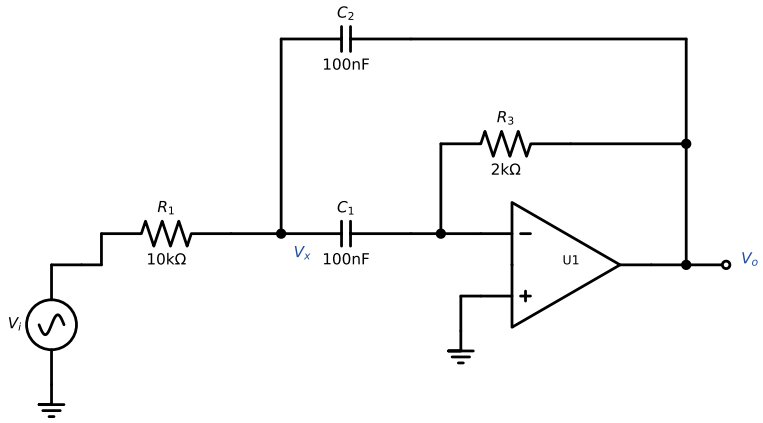
Problem 2B.8 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 30000\,t)$.



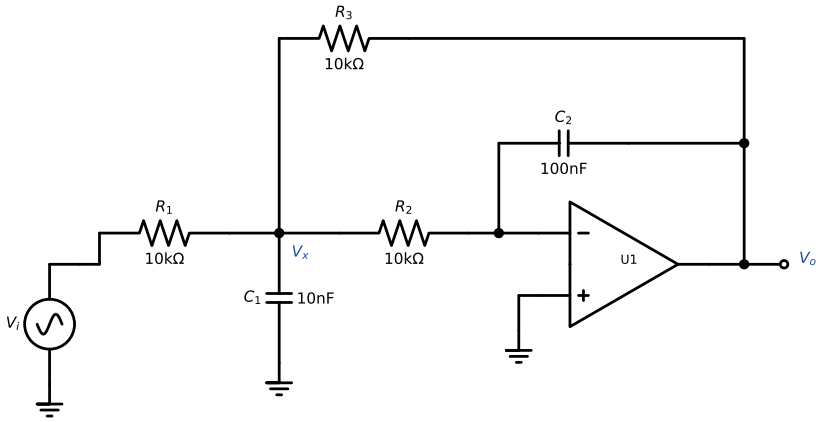
Problem 2B.9 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1\sin(2\pi \cdot 300\,t)$.



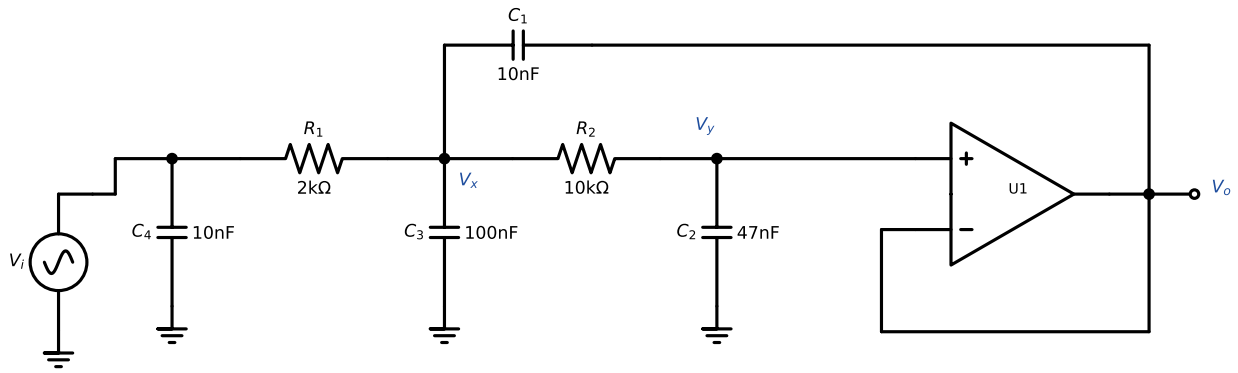
Problem 2B.10 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 200\,t)$.



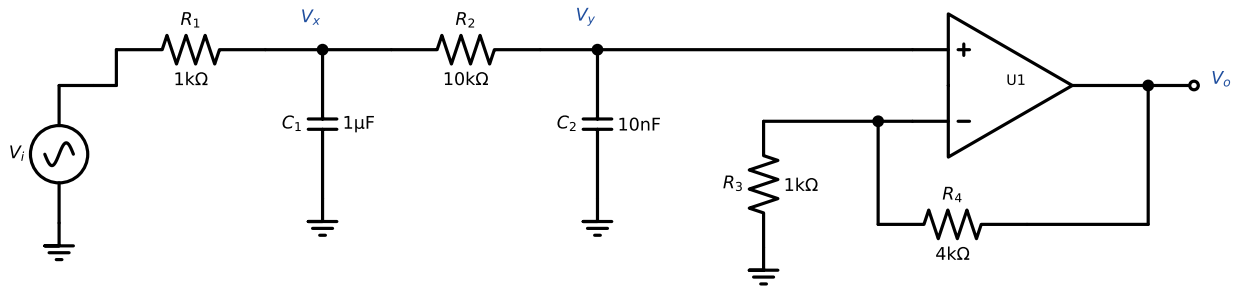
Problem 2B.11 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin\left(2\pi \cdot 500\,t + \frac{\pi}{4}\right)$.



Problem 2B.12 — Medium

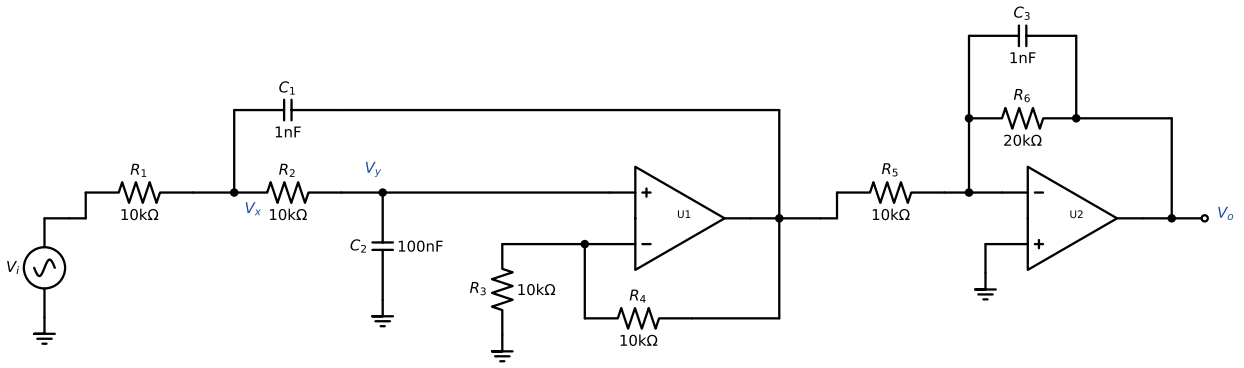
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin\left(2\pi \cdot 1000\,t\right)$.



Hard problems

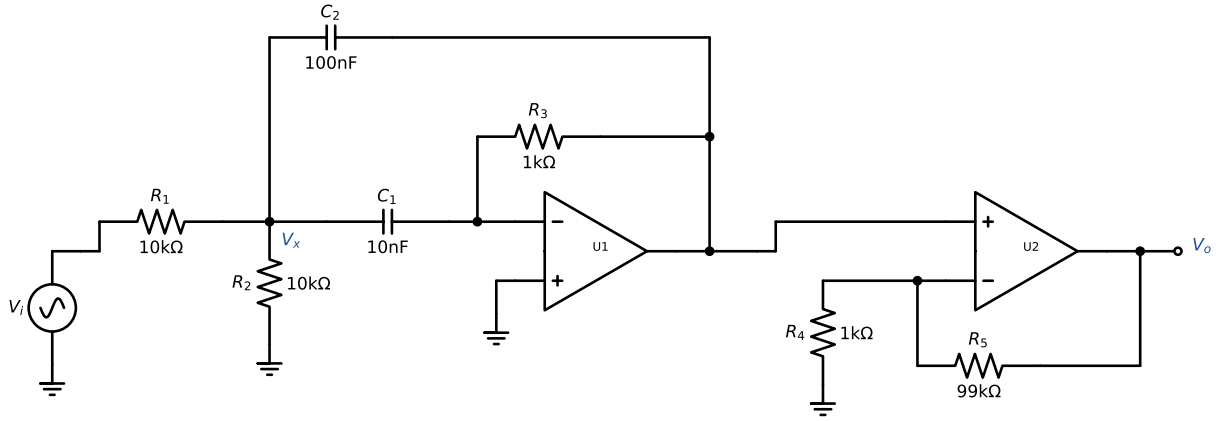
Problem 2B.13 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$.



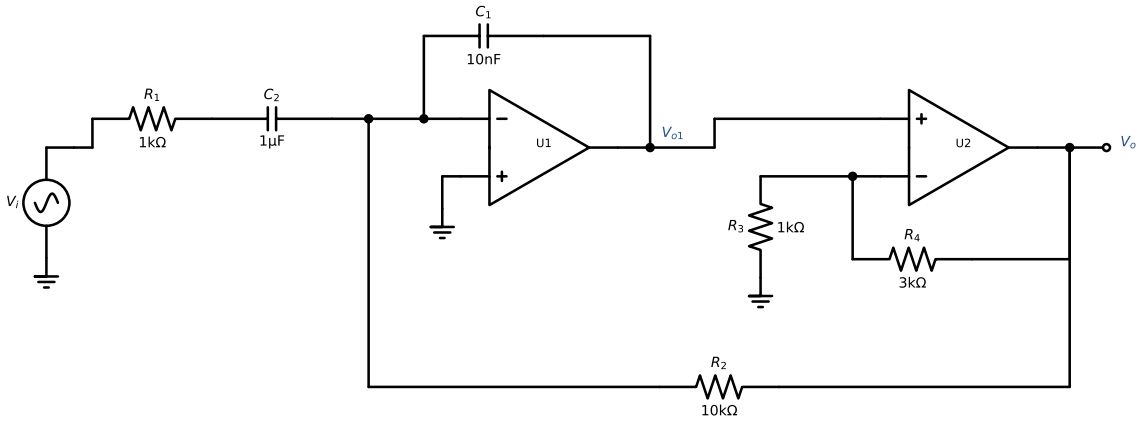
Problem 2B.14 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$.



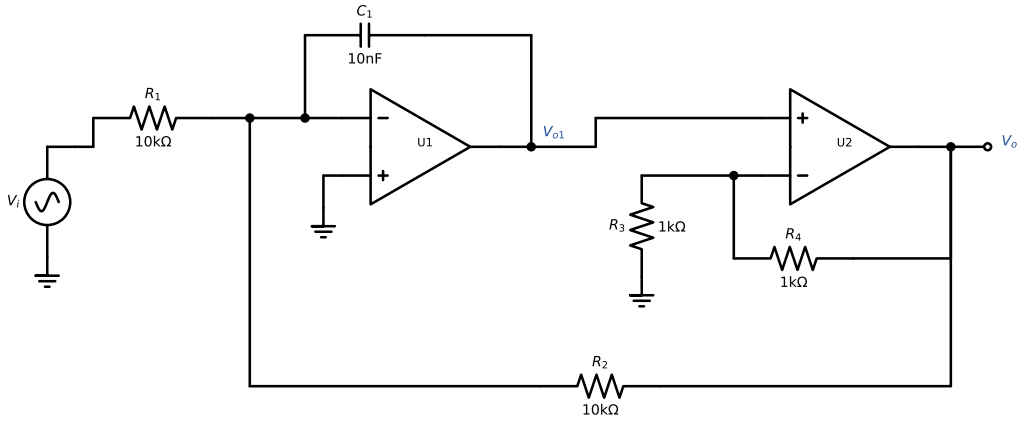
Problem 2B.15 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$.



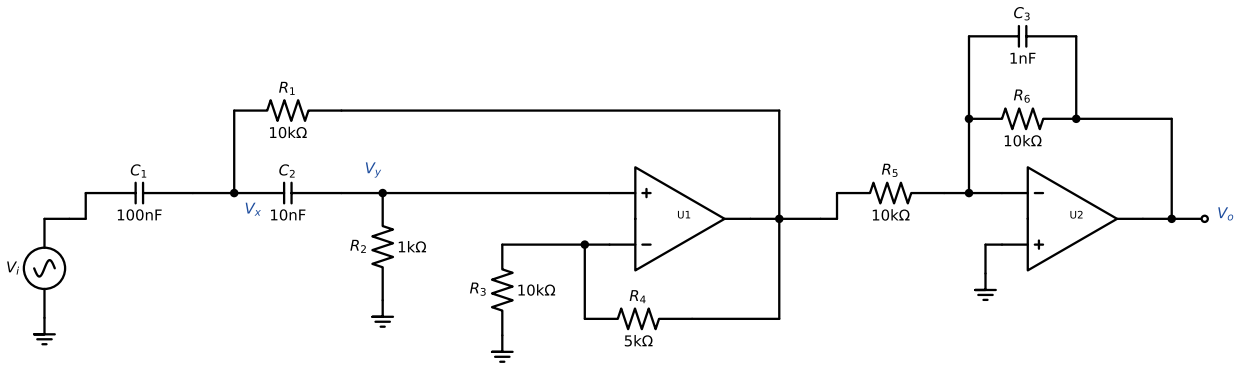
Problem 2B.16 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\cos\left(2\pi \cdot 5000\,t - \frac{\pi}{4}\right)$.



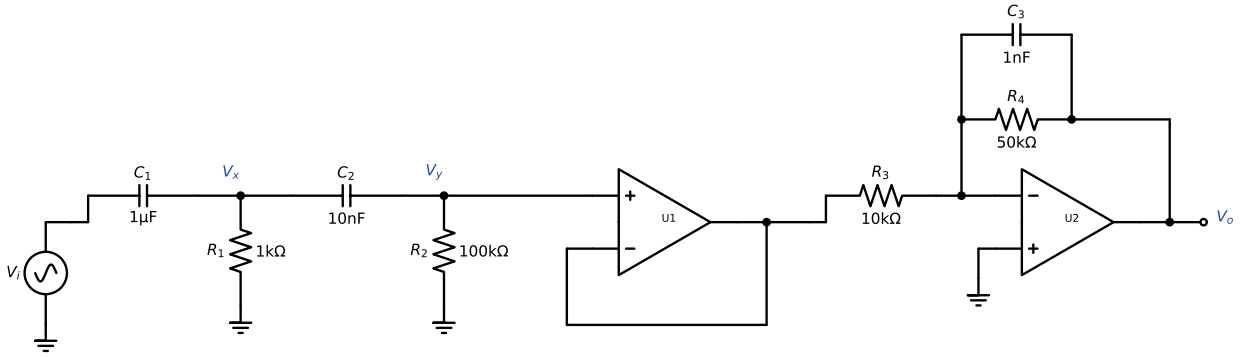
Problem 2B.17 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 20000\,t)$.



Problem 2B.18 — Hard

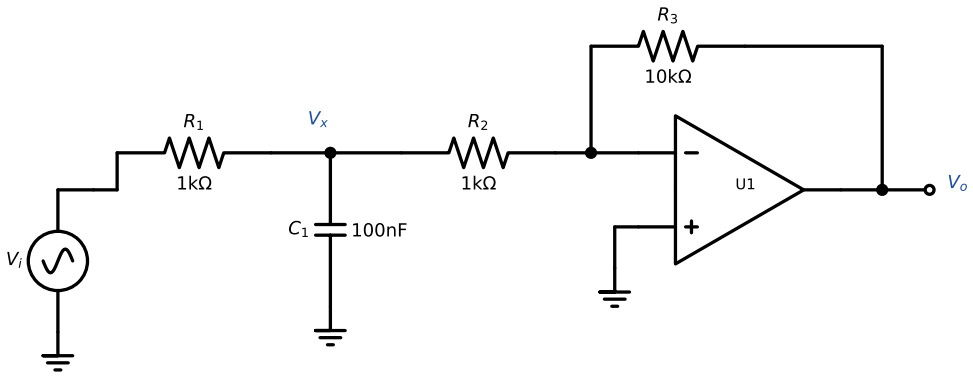
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin\left(2\pi \cdot 1000\,t + \frac{\pi}{3}\right)$.



Solutions

Every solution follows the steps of the Lecture-2 pipeline card.

Solution 2B.1 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Inverting amplifier whose input network has a shunt branch. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (=0): \qquad \frac{0 - V_x}{R_2} + \frac{0 - V_o}{R_3} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + sC_1 V_x + \frac{V_x}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (2):} \quad V_x = \left(\frac{10000}{s + 20000}\right) V_i$$

Substituting everything into (1) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-10^5}{s + 20000}$$

So:

$$H(s) = \frac{-10^5}{s + 20000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 20000$; $s + 20000 = 0 \Rightarrow s = -20000$.

Factored form (leading coefficient in front):

$$H(s) = -10^5 \, \frac{1}{(s + 20000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}}, \qquad K = -5$$

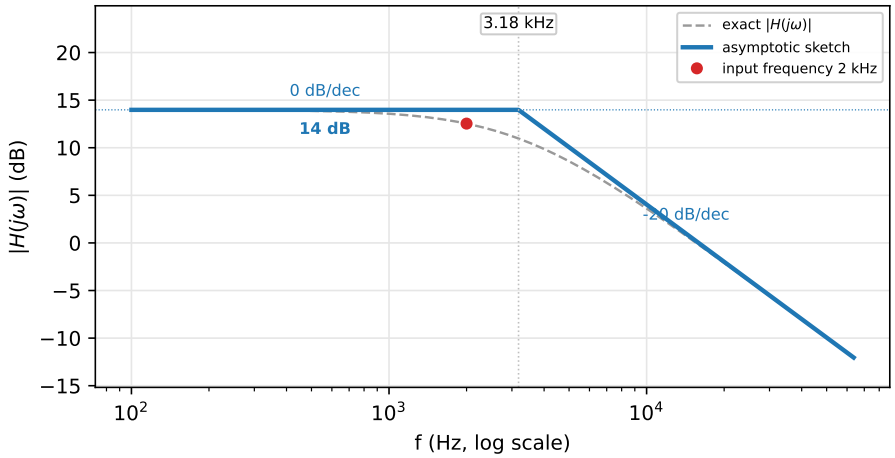
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	20000	3.18 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.18$ kHz	0 dB/dec (flat)	14 dB at end
$f > 3.18$ kHz	−20 dB/dec	14 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}}$	0.8467	-0.561
total	**4.234** (product)	$-$**3.7** (sum)

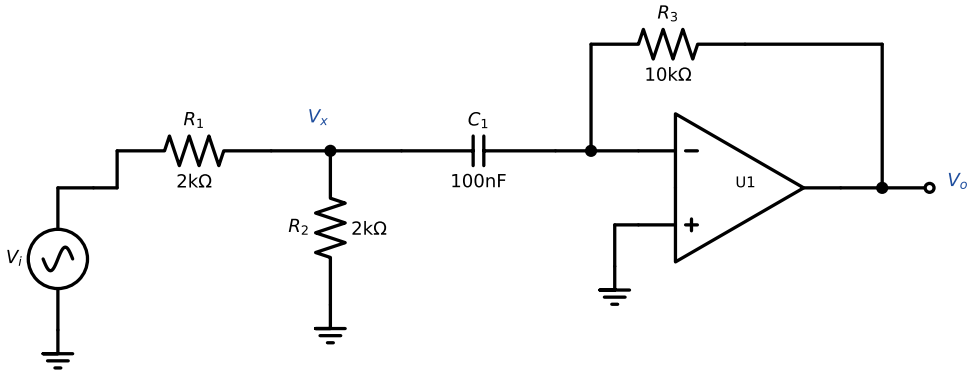
$$|H(j\omega_{in})| = 4.23, \qquad \angle H(j\omega_{in}) = -3.7 \text{ rad}$$
$$v_{o,ac}(t) = 4.23 \times 0.1 \sin(2\pi \cdot 2000\,t - 3.7) = 0.423 \sin(2\pi \cdot 2000\,t - 3.7)$$

$$\boxed{V_o(t) = 0.423 \sin(2\pi \cdot 2000\,t - 3.7) \text{ V}}$$

Answer

$$H(s) = -5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}} \quad \text{(corners: 3.18 kHz)}$$
$$V_o(t) = 0.423 \sin(2\pi \cdot 2000\,t - 3.7) \text{ V}$$

Solution 2B.2 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Inverting amplifier whose input network has a shunt branch. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (=0): \qquad sC_1\,(0 - V_x) + \frac{0 - V_o}{R_3} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + \frac{V_x}{R_2} + sC_1 V_x = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (2):} \quad V_x = \left(\frac{5000}{s + 10000}\right) V_i$$

Substituting everything into (1) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-5\,s}{s + 10000}$$

So:

$$H(s) = \frac{-5\,s}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = -5\,\frac{s}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -5\,\frac{s}{s + 2\pi \cdot 1.59\text{k}}, \qquad K = -5$$

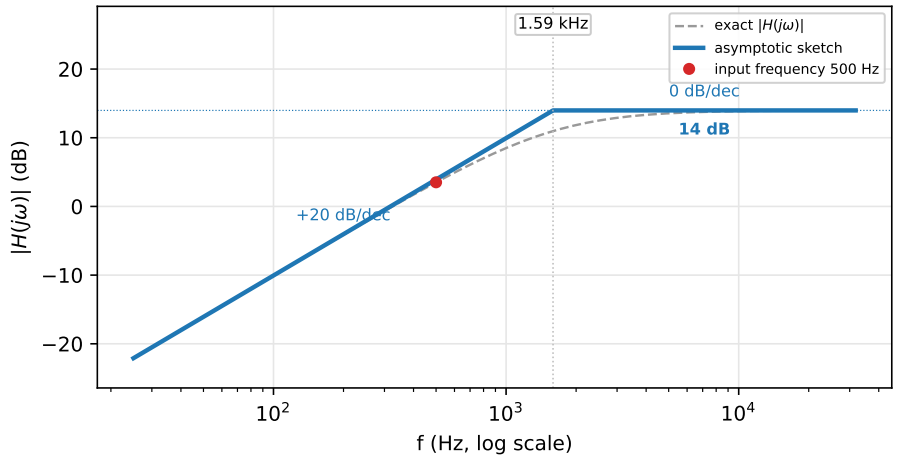
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 5 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	+20 dB/dec	14 dB at end
$f > 1.59$ kHz	0 dB/dec (flat)	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{s}{s + 2\pi \cdot 1.59\text{k}}$	0.2997	1.27
total	**1.499** (product)	$-$**1.88** (sum)

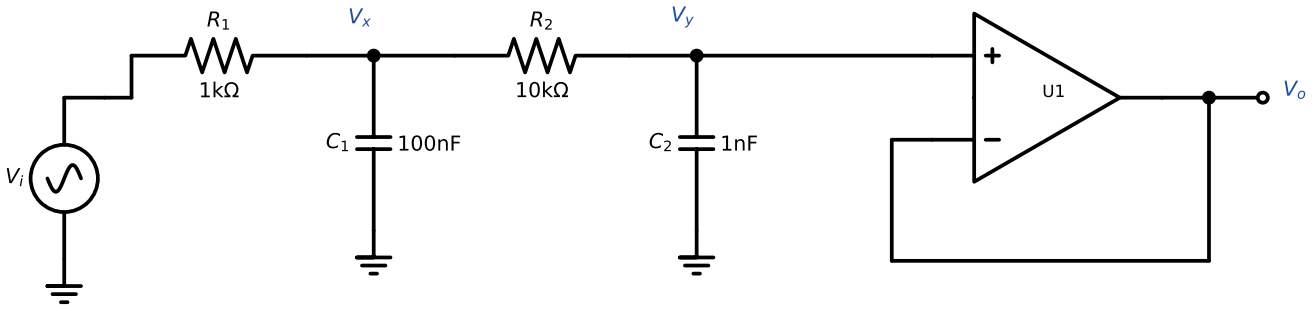
$$|H(j\omega_{in})| = 1.5, \qquad \angle H(j\omega_{in}) = -1.88 \text{ rad}$$
$$v_{o,ac}(t) = 1.5 \times 0.2 \sin(2\pi \cdot 500\,t - 1.88) = 0.3 \sin(2\pi \cdot 500\,t - 1.88)$$

$$\boxed{V_o(t) = 0.3 \sin(2\pi \cdot 500\,t - 1.88) \text{ V}}$$

Answer

$$H(s) = -5\,\frac{s}{s + 2\pi \cdot 1.59\text{k}} \quad \text{(corners: 1.59 kHz)}$$
$$V_o(t) = 0.3 \sin(2\pi \cdot 500\,t - 1.88) \text{ V}$$

Solution 2B.3 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). A two-section passive ladder drives the + input. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_o is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_x: \qquad \frac{V_x - V_i}{R_1} + sC_1V_x + \frac{V_x - V_y}{R_2} = 0$$

$$\text{(2) at } V_y: \qquad \frac{V_y - V_x}{R_2} + sC_2V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{1000}{s+11000}\right)V_y + \left(\frac{10000}{s+11000}\right)V_i$$

Substituting everything into (2) leaves only V_y and V_i. Since $V_o = V_y$ (virtual short),

$$\frac{V_o}{V_i} = \frac{V_y}{V_i} = \frac{10^9}{s^2 + 1.11\times10^5\, s + 10^9}$$

So:

$$H(s) = \frac{10^9}{s^2 + 1.11\times10^5\, s + 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 1.11 \times 10^5\, s + 10^9$; quadratic formula:

$$s = \frac{-1.11 \times 10^5 \pm \sqrt{(1.11 \times 10^5)^2 - 4(10^9)}}{2} = \frac{-1.11 \times 10^5 \pm 91200}{2} \Rightarrow s = -9890,\ -1.01 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 10^9 \frac{1}{(s + 9890)(s + 1.01 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1 \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.57\text{k}}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 16.1\text{k}}}, \qquad K = 1$$

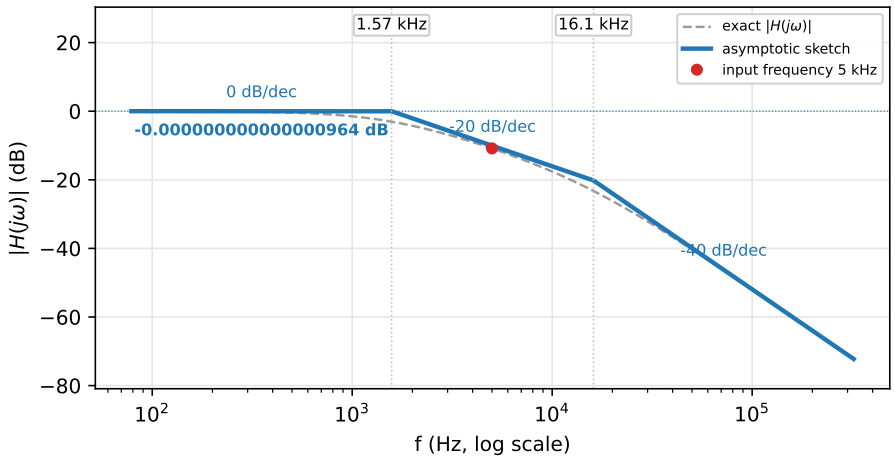
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	9890	1.57 kHz	-20 dB/dec
pole	1.01×10^5	16.1 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.00000000000000000964$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.57$ kHz	0 dB/dec (flat)	0 dB at end
1.57 kHz – 16.1 kHz	-20 dB/dec	0 dB at start, -20.2 dB at end
$f > 16.1$ kHz	-40 dB/dec	-20.2 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = -0.00000000000000000964$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.57\text{k}}}$	0.3003	-1.27
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 16.1\text{k}}}$	0.955	-0.301
total	**0.2868** (product)	$\mathbf{-1.57}$ (sum)

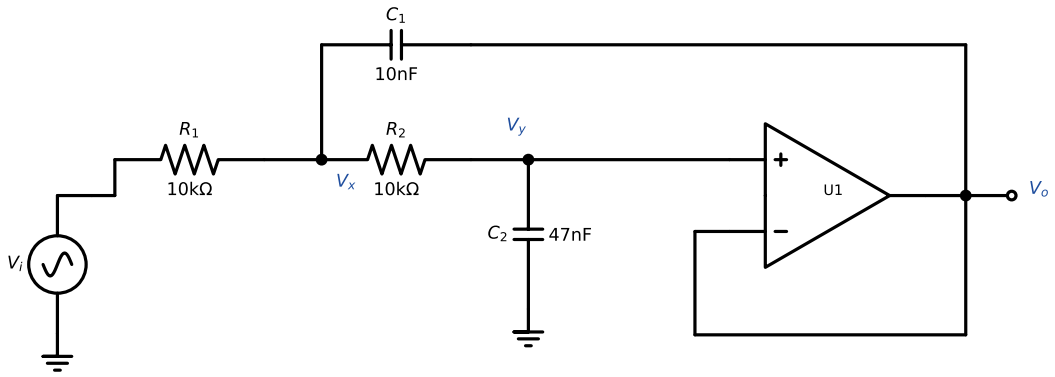
$$|H(j\omega_{in})| = 0.287, \qquad \angle H(j\omega_{in}) = -1.57 \text{ rad}$$
$$v_{o,ac}(t) = 0.287 \times 1 \sin(2\pi \cdot 5000\,t - 1.57) = 0.287 \sin(2\pi \cdot 5000\,t - 1.57)$$

$$\boxed{V_o(t) = 0.287 \sin(2\pi \cdot 5000\,t - 1.57)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.57\text{k}}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 16.1\text{k}}} \quad \text{(corners: 1.57 kHz, 16.1 kHz)}$$
$$V_o(t) = 0.287 \sin(2\pi \cdot 5000\,t - 1.57)\ \text{V}$$

Solution 2B.4 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_o is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_x: \qquad \frac{V_x - V_i}{R_1} + \frac{V_x - V_y}{R_2} + sC_1\,(V_x - V_y) = 0$$

$$\text{(2) at } V_y: \qquad \frac{V_y - V_x}{R_2} + sC_2 V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{s + 10000}{s + 20000}\right) V_y + \left(\frac{10000}{s + 20000}\right) V_i$$

Substituting everything into (2) leaves only V_y and V_i. Since $V_o = V_y$ (virtual short),

$$\frac{V_o}{V_i} = \frac{V_y}{V_i} = \frac{2.13 \times 10^7}{s^2 + 20000\,s + 2.13 \times 10^7}$$

So:

$$H(s) = \frac{2.13 \times 10^7}{s^2 + 20000\,s + 2.13 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 20000\,s + 2.13 \times 10^7$; quadratic formula:

$$s = \frac{-20000 \pm \sqrt{(20000)^2 - 4(2.13 \times 10^7)}}{2} = \frac{-20000 \pm 17700}{2} \Rightarrow s = -1130,\ -18900$$

Factored form (leading coefficient in front):

$$H(s) = 2.13 \times 10^7 \, \frac{1}{(s + 1130)(s + 18900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 179}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 3\text{k}}}, \qquad K = 1$$

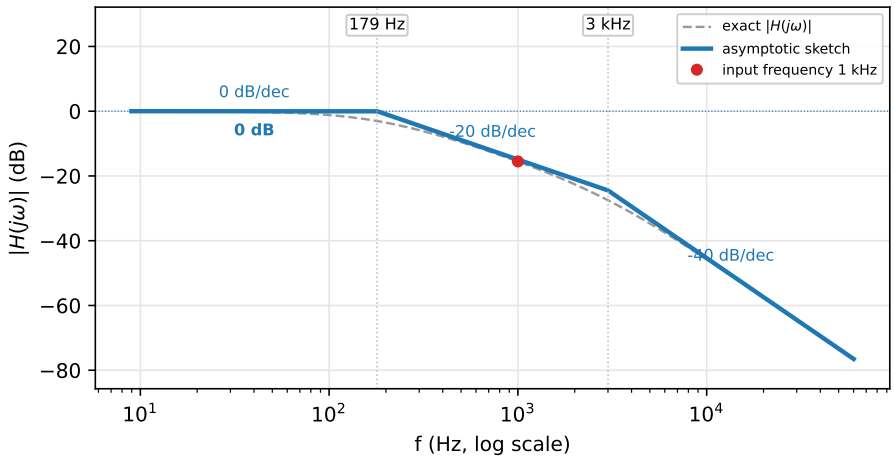
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1130	179 Hz	−20 dB/dec
pole	18900	3 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 179$ Hz	0 dB/dec (flat)	0 dB at end
179 Hz – 3 kHz	−20 dB/dec	0 dB at start, -24.5 dB at end
$f > 3$ kHz	−40 dB/dec	-24.5 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 179}}$	0.1766	-1.39
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 3\text{k}}}$	0.9488	-0.321
total	**0.1676** (product)	**−1.71** (sum)

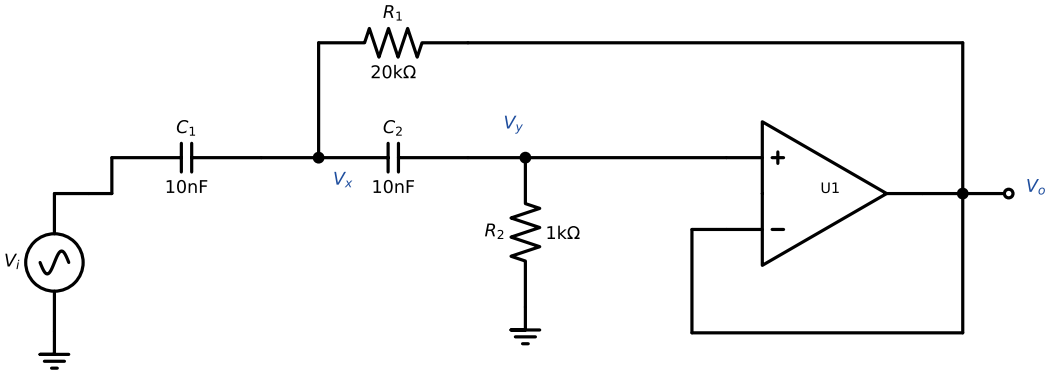
$$|H(j\omega_{in})| = 0.168, \qquad \angle H(j\omega_{in}) = -1.71 \text{ rad}$$
$$v_{o,ac}(t) = 0.168 \times 0.5\sin(2\pi \cdot 1000\,t - 1.71) = 0.0838\sin(2\pi \cdot 1000\,t - 1.71)$$

$$\boxed{V_o(t) = 0.0838\sin(2\pi \cdot 1000\,t - 1.71)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 179}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 3\text{k}}} \quad \text{(corners: 179 Hz, 3 kHz)}$$
$$V_o(t) = 0.0838\sin(2\pi \cdot 1000\,t - 1.71)\ \text{V}$$

Solution 2B.5 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_o is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_x: \qquad sC_1\,(V_x - V_i) + sC_2\,(V_x - V_y) + \frac{V_x - V_y}{R_1} = 0$$

$$\text{(2) at } V_y: \qquad sC_2\,(V_y - V_x) + \frac{V_y}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{0.5\,s + 2500}{s + 2500}\right) V_y + \left(\frac{0.5\,s}{s + 2500}\right) V_i$$

Substituting everything into (2) leaves only V_y and V_i. Since $V_o = V_y$ (virtual short),

$$\frac{V_o}{V_i} = \frac{V_y}{V_i} = \frac{s^2}{s^2 + 2 \times 10^5\,s + 5 \times 10^8}$$

So:

$$H(s) = \frac{s^2}{s^2 + 2 \times 10^5\,s + 5 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^2 + 2 \times 10^5\, s + 5 \times 10^8$; quadratic formula:

$$s = \frac{-2 \times 10^5 \pm \sqrt{(2 \times 10^5)^2 - 4(5 \times 10^8)}}{2} = \frac{-2 \times 10^5 \pm 1.95 \times 10^5}{2} \Rightarrow s = -2530,\ -1.97 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 1\, \frac{s^2}{(s + 2530)(s + 1.97 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\, \frac{s}{s + 2\pi \cdot 403}\, \frac{s}{s + 2\pi \cdot 31.4\text{k}}, \qquad K = 1$$

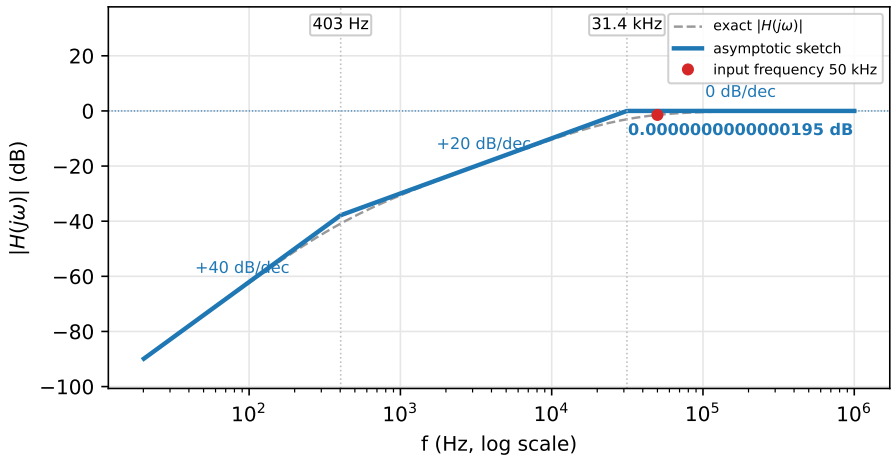
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2530	403 Hz	-20 dB/dec
pole	1.97×10^5	31.4 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 2 \times 10^{-9}\, s^2$, so the plot starts with a slope of $+40$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 403 Hz	+40 dB/dec	-37.8 dB at end
403 Hz – 31.4 kHz	+20 dB/dec	-37.8 dB at start, 0 dB at end
$f >$ 31.4 kHz	0 dB/dec (flat)	0 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = 0$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 50000\,t)$, so $\omega_{in} = 2\pi{\cdot}50000 = 3.14{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{s}{s+2\pi\cdot 403}$	1	0.00806
$\dfrac{s}{s+2\pi\cdot 31.4\text{k}}$	0.8466	0.561
total	**0.8466** (product)	**0.569** (sum)

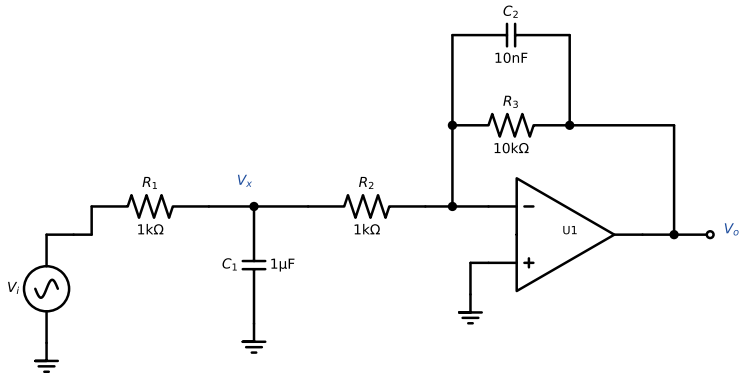
$$|H(j\omega_{in})| = 0.847, \qquad \angle H(j\omega_{in}) = 0.569 \text{ rad}$$
$$v_{o,ac}(t) = 0.847 \times 0.5 \sin(2\pi\cdot 50000\,t + 0.569) = 0.423 \sin(2\pi\cdot 50000\,t + 0.569)$$

$$\boxed{V_o(t) = 0.423 \sin(2\pi\cdot 50000\,t + 0.569)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{s}{s+2\pi\cdot 403}\,\frac{s}{s+2\pi\cdot 31.4\text{k}} \quad \text{(corners: 403 Hz, 31.4 kHz)}$$
$$V_o(t) = 0.423 \sin(2\pi\cdot 50000\,t + 0.569)\ \text{V}$$

Solution 2B.6 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Inverting amplifier whose input network has a shunt branch. This is not a standard block, so use nodal analysis. Short names for the composite branches:

$$Z_f = R_3 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 10000}$$

Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= 0): \qquad \frac{0 - V_x}{R_2} + \frac{0 - V_o}{Z_f} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + sC_1 V_x + \frac{V_x}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (2):} \quad V_x = \left(\frac{1000}{s + 2000}\right) V_i$$

Substituting everything into (1) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-10^8}{s^2 + 12000\, s + 2 \times 10^7}$$

So:

$$H(s) = \frac{-10^8}{s^2 + 12000\,s + 2 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 12000\,s + 2 \times 10^7$; quadratic formula:

$$s = \frac{-12000 \pm \sqrt{(12000)^2 - 4(2 \times 10^7)}}{2} = \frac{-12000 \pm 8000}{2} \Rightarrow s = -2000,\ -10000$$

Factored form (leading coefficient in front):

$$H(s) = -10^8\,\frac{1}{(s + 2000)(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -5\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 318}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = -5$$

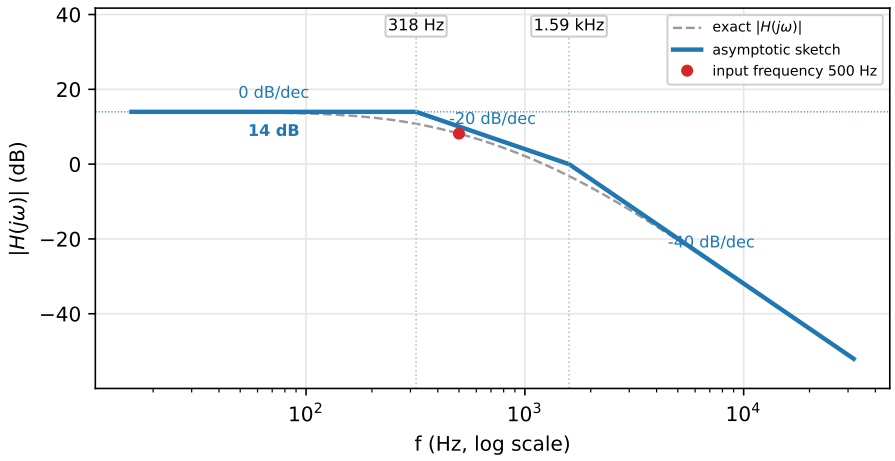
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2000	318 Hz	−20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 318$ Hz	0 dB/dec (flat)	14 dB at end
318 Hz – 1.59 kHz	−20 dB/dec	14 dB at start, 0 dB at end
$f > 1.59$ kHz	−40 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\cos(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 318}}$	0.537	-1
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.954	-0.304
total	**2.562** (product)	$-$**4.45** (sum)

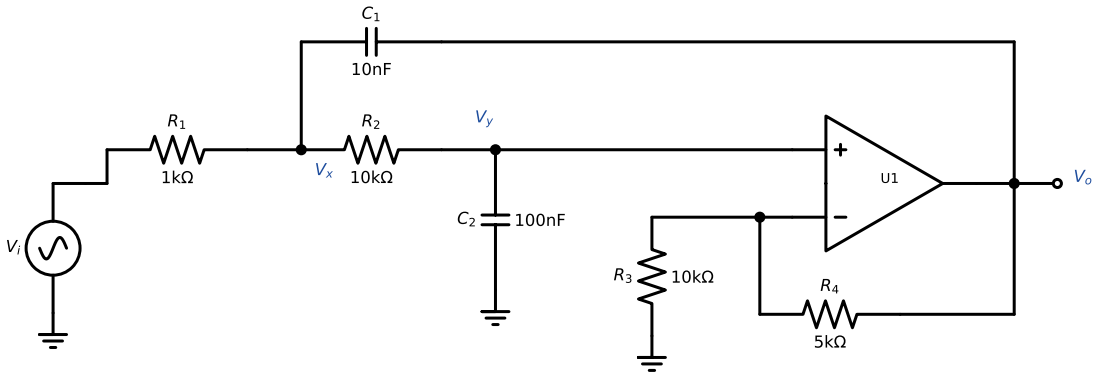
$$|H(j\omega_{in})| = 2.56, \qquad \angle H(j\omega_{in}) = -4.45 \text{ rad}$$
$$v_{o,ac}(t) = 2.56 \times 0.1\cos(2\pi \cdot 500\,t - 4.45) = 0.256\cos(2\pi \cdot 500\,t - 4.45)$$

$$\boxed{V_o(t) = 0.256\cos(2\pi \cdot 500\,t - 4.45)\ \text{V}}$$

Answer

$$H(s) = -5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 318}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}} \quad \text{(corners: 318 Hz, 1.59 kHz)}$$
$$V_o(t) = 0.256\cos(2\pi \cdot 500\,t - 4.45)\ \text{V}$$

Solution 2B.7 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= V_y): \qquad \frac{V_y}{R_3} + \frac{V_y - V_o}{R_4} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + \frac{V_x - V_y}{R_2} + sC_1 (V_x - V_o) = 0$$

$$\text{(3) at } V_y: \qquad \frac{V_y - V_x}{R_2} + sC_2 V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_y = 0.667\, V_o$$

$$\text{from (2):} \quad V_x = \left(\frac{s + 6670}{s + 1.1 \times 10^5}\right) V_o + \left(\frac{10^5}{s + 1.1 \times 10^5}\right) V_i$$

Substituting everything into (3) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{1.5 \times 10^8}{s^2 + 1.09 \times 10^5\, s + 10^8}$$

So:

$$H(s) = \frac{1.5 \times 10^8}{s^2 + 1.09 \times 10^5\, s + 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 1.09 \times 10^5\, s + 10^8$; quadratic formula:

$$s = \frac{-1.09 \times 10^5 \pm \sqrt{(1.09 \times 10^5)^2 - 4(10^8)}}{2} = \frac{-1.09 \times 10^5 \pm 1.08 \times 10^5}{2} \Rightarrow s = -921,\ -1.09 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 1.5 \times 10^8 \, \frac{1}{(s + 921)(s + 1.09 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1.5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 147}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 17.3\text{k}}}, \qquad K = 1.5$$

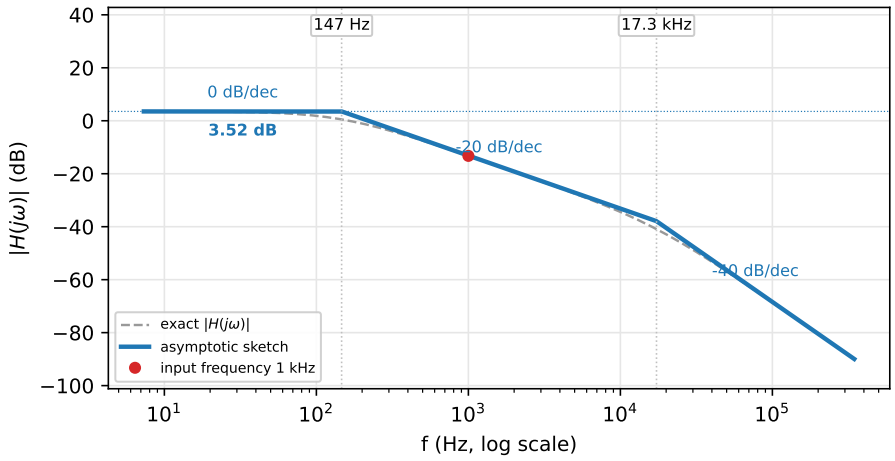
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	921	147 Hz	−20 dB/dec
pole	1.09×10^5	17.3 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1.5 = 3.52\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 147$ Hz	0 dB/dec (flat)	3.52 dB at end
147 Hz – 17.3 kHz	−20 dB/dec	3.52 dB at start, -37.9 dB at end
$f > 17.3$ kHz	−40 dB/dec	-37.9 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1.5 = 3.52\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.5	1.5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 147}}$	0.145	-1.43
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 17.3\text{k}}}$	0.9983	-0.0578
total	**0.2172** (product)	**−1.48** (sum)

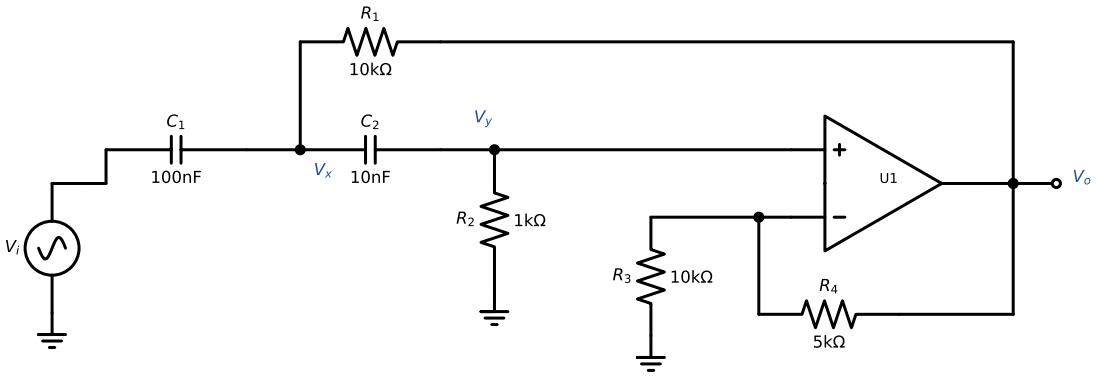
$$|H(j\omega_{in})| = 0.217, \qquad \angle H(j\omega_{in}) = -1.48 \text{ rad}$$
$$v_{o,ac}(t) = 0.217 \times 0.2\sin(2\pi \cdot 1000\,t - 1.48) = 0.0434\sin(2\pi \cdot 1000\,t - 1.48)$$

$$\boxed{V_o(t) = 0.0434\sin(2\pi \cdot 1000\,t - 1.48)\ \text{V}}$$

Answer

$$H(s) = 1.5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 147}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 17.3\text{k}}} \quad \text{(corners: 147 Hz, 17.3 kHz)}$$
$$V_o(t) = 0.0434\sin(2\pi \cdot 1000\,t - 1.48)\ \text{V}$$

Solution 2B.8 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= V_y): \qquad \frac{V_y}{R_3} + \frac{V_y - V_o}{R_4} = 0$$

$$\text{(2) at } V_x: \qquad sC_1 (V_x - V_i) + sC_2 (V_x - V_y) + \frac{V_x - V_o}{R_1} = 0$$

$$\text{(3) at } V_y: \qquad sC_2 (V_y - V_x) + \frac{V_y}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_y = 0.667\, V_o$$

$$\text{from (2):} \quad V_x = \left(\frac{0.0606\, s + 909}{s + 909}\right) V_o + \left(\frac{0.909\, s}{s + 909}\right) V_i$$

Substituting everything into (3) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{1.5\, s^2}{s^2 + 1.09 \times 10^5\, s + 10^8}$$

So:

$$H(s) = \frac{1.5\, s^2}{s^2 + 1.09 \times 10^5\, s + 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow 2$ zeros at $s = 0$.
- Denominator: $s^2 + 1.09 \times 10^5\, s + 10^8$; quadratic formula:

$$s = \frac{-1.09 \times 10^5 \pm \sqrt{(1.09 \times 10^5)^2 - 4(10^8)}}{2} = \frac{-1.09 \times 10^5 \pm 1.08 \times 10^5}{2} \Rightarrow s = -921,\ -1.09 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = 1.5\, \frac{s^2}{(s + 921)(s + 1.09 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1.5\, \frac{s}{s + 2\pi \cdot 147}\, \frac{s}{s + 2\pi \cdot 17.3\text{k}}, \qquad K = 1.5$$

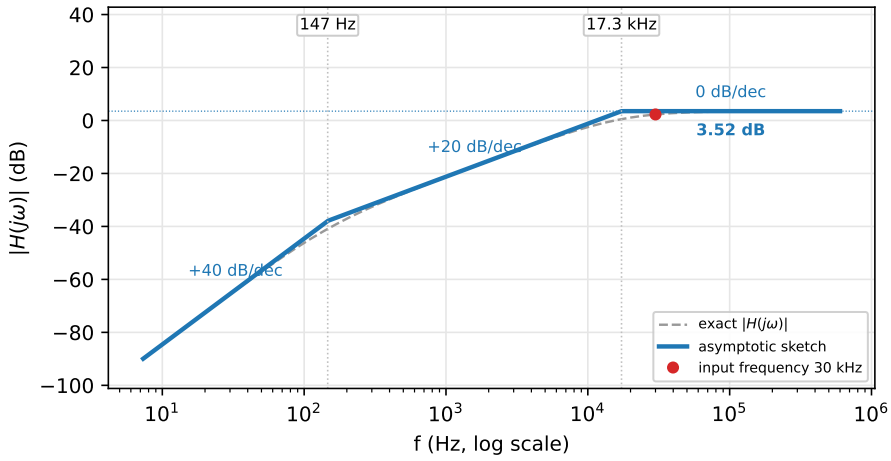
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	921	147 Hz	−20 dB/dec
pole	1.09×10^5	17.3 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.5 \times 10^{-8}\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 147 Hz	+40 dB/dec	-37.9 dB at end
147 Hz – 17.3 kHz	+20 dB/dec	-37.9 dB at start, 3.52 dB at end
$f >$ 17.3 kHz	0 dB/dec (flat)	3.52 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1.5 = 3.52$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 30000\,t)$, so $\omega_{in} = 2\pi{\cdot}30000 = 1.88{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1.5	1.5	0
$\dfrac{s}{s+2\pi\cdot 147}$	1	0.00489
$\dfrac{s}{s+2\pi\cdot 17.3\text{k}}$	0.8665	0.523
total	**1.3** (product)	**0.527** (sum)

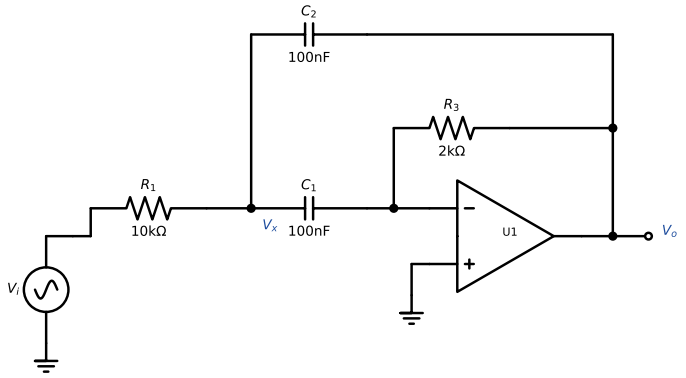
$$|H(j\omega_{in})| = 1.3, \qquad \angle H(j\omega_{in}) = 0.527 \text{ rad}$$
$$v_{o,ac}(t) = 1.3 \times 0.2\,\sin(2\pi\cdot 30000\,t + 0.527) = 0.26\,\sin(2\pi\cdot 30000\,t + 0.527)$$

$$\boxed{V_o(t) = 0.26\,\sin(2\pi\cdot 30000\,t + 0.527)\ \text{V}}$$

Answer

$$H(s) = 1.5\,\frac{s}{s+2\pi\cdot 147}\,\frac{s}{s+2\pi\cdot 17.3\text{k}} \quad \text{(corners: 147 Hz, 17.3 kHz)}$$
$$V_o(t) = 0.26\,\sin(2\pi\cdot 30000\,t + 0.527)\ \text{V}$$

Solution 2B.9 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Multiple-feedback stage: node V_x is fed back from the output through a second branch. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (=0): \qquad sC_1(0 - V_x) + \frac{0 - V_o}{R_3} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + sC_1V_x + sC_2(V_x - V_o) = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{-5000}{s}\right) V_o$$

Substituting everything into (2) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-1000\,s}{s^2 + 10000\,s + 5 \times 10^6}$$

So:

$$H(s) = \frac{-1000\,s}{s^2 + 10000\,s + 5 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 10000\,s + 5 \times 10^6$; quadratic formula:

$$s = \frac{-10000 \pm \sqrt{(10000)^2 - 4(5 \times 10^6)}}{2} = \frac{-10000 \pm 8940}{2} \Rightarrow s = -528,\ -9470$$

Factored form (leading coefficient in front):

$$H(s) = -1000\,\frac{s}{(s+528)(s+9470)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -0.106\,\frac{s}{s + 2\pi \cdot 84}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.51\text{k}}}, \qquad K = -0.106$$

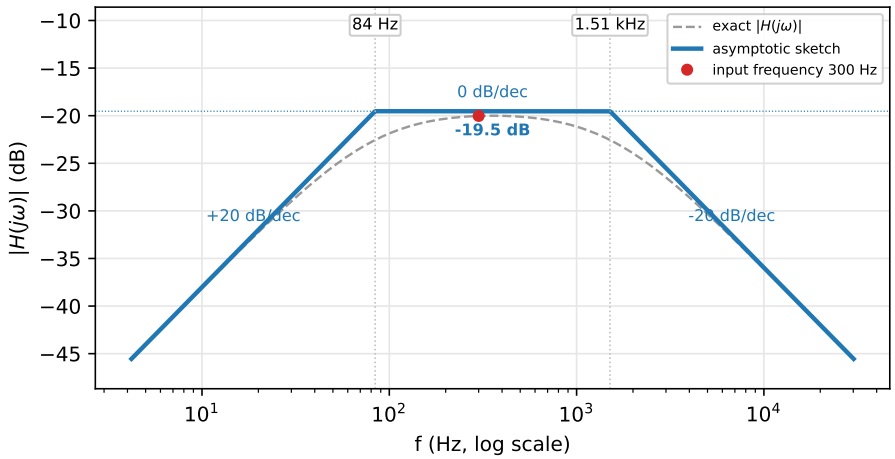
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	528	84 Hz	-20 dB/dec
pole	9470	1.51 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 2 \times 10^{-4}\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 84$ Hz	$+20$ dB/dec	-19.5 dB at end
84 Hz – 1.51 kHz	0 dB/dec (flat)	-19.5 dB at start, -19.5 dB at end
$f > 1.51$ kHz	-20 dB/dec	-19.5 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.106 = -19.5$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.106	0.1056	-3.14
$\dfrac{s}{s+2\pi\cdot 84}$	0.963	0.273
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.51\text{k}}}$	0.9808	-0.196
total	**0.09971** (product)	$-$**3.06** (sum)

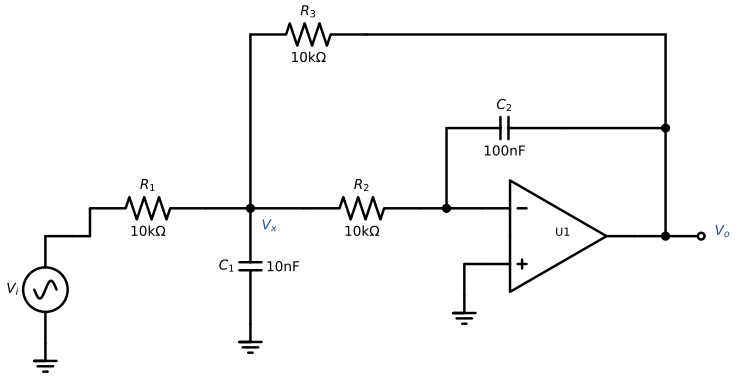
$$|H(j\omega_{in})| = 0.0997, \qquad \angle H(j\omega_{in}) = -3.06 \text{ rad}$$
$$v_{o,ac}(t) = 0.0997 \times 1 \sin(2\pi \cdot 300\,t - 3.06) = 0.0997 \sin(2\pi \cdot 300\,t - 3.06)$$

$$\boxed{V_o(t) = 0.0997 \sin(2\pi \cdot 300\,t - 3.06)\ \text{V}}$$

Answer

$$H(s) = -0.106\,\frac{s}{s+2\pi\cdot 84}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.51\text{k}}} \quad \text{(corners: 84 Hz, 1.51 kHz)}$$
$$V_o(t) = 0.0997 \sin(2\pi \cdot 300\,t - 3.06)\ \text{V}$$

Solution 2B.10 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Multiple-feedback stage: node V_x is fed back from the output through a second branch. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (=0): \qquad \frac{0 - V_x}{R_2} + sC_2\,(0 - V_o) = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + \frac{V_x}{R_2} + \frac{V_x - V_o}{R_3} + sC_1 V_x = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = -0.001\,s\,V_o$$

Substituting everything into (2) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-10^7}{s^2 + 30000\,s + 10^7}$$

So:

$$H(s) = \frac{-10^7}{s^2 + 30000\,s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 30000\,s + 10^7$; quadratic formula:

$$s = \frac{-30000 \pm \sqrt{(30000)^2 - 4(10^7)}}{2} = \frac{-30000 \pm 29300}{2} \Rightarrow s = -337,\ -29700$$

Factored form (leading coefficient in front):

$$H(s) = -10^7\,\frac{1}{(s+337)(s+29700)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 53.7}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 4.72\text{k}}}, \qquad K = -1$$

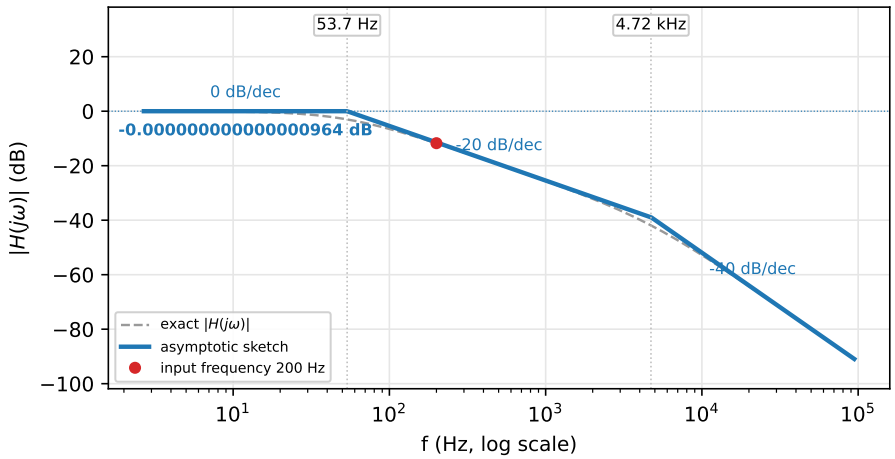
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	337	53.7 Hz	−20 dB/dec
pole	29700	4.72 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.000000000000000964$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 53.7$ Hz	0 dB/dec (flat)	0 dB at end
53.7 Hz – 4.72 kHz	−20 dB/dec	0 dB at start, -38.9 dB at end
$f > 4.72$ kHz	−40 dB/dec	-38.9 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = -0.000000000000000964$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 200\,t)$, so $\omega_{in} = 2\pi \cdot 200 = 1260$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1	1	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 53.7}}$	0.2591	-1.31
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 4.72\text{k}}}$	0.9991	-0.0423
total	**0.2589** (product)	$\mathbf{-4.49}$ (sum)

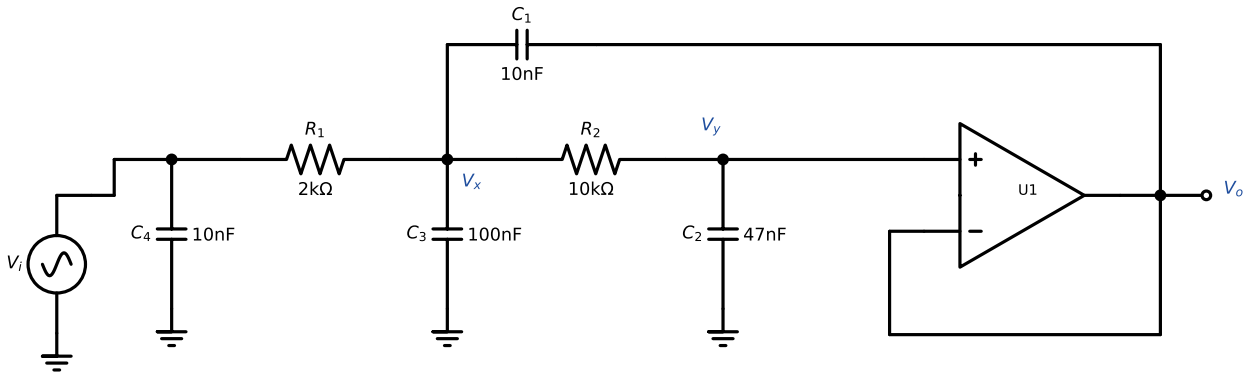
$$|H(j\omega_{in})| = 0.259, \qquad \angle H(j\omega_{in}) = -4.49 \text{ rad}$$
$$v_{o,ac}(t) = 0.259 \times 0.5\,\sin(2\pi \cdot 200\,t - 4.49) = 0.129\,\sin(2\pi \cdot 200\,t - 4.49)$$

$$\boxed{V_o(t) = 0.129\,\sin(2\pi \cdot 200\,t - 4.49)\ \text{V}}$$

Answer

$$H(s) = -1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 53.7}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 4.72\text{k}}} \quad \text{(corners: 53.7 Hz, 4.72 kHz)}$$
$$V_o(t) = 0.129\,\sin(2\pi \cdot 200\,t - 4.49)\ \text{V}$$

Solution 2B.11 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_o is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_x: \qquad \frac{V_x - V_i}{R_1} + \frac{V_x - V_y}{R_2} + sC_1\left(V_x - V_y\right) + sC_3V_x = 0$$

$$\text{(2) at } V_y: \qquad \frac{V_y - V_x}{R_2} + sC_2V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{0.0909\,s + 909}{s + 5450}\right)V_y + \left(\frac{4550}{s + 5450}\right)V_i$$

Substituting everything into (2) leaves only V_y and V_i. Since $V_o = V_y$ (virtual short),

$$\frac{V_o}{V_i} = \frac{V_y}{V_i} = \frac{9.67 \times 10^6}{s^2 + 7390\,s + 9.67 \times 10^6}$$

So:

$$H(s) = \frac{9.67 \times 10^6}{s^2 + 7390\,s + 9.67 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 7390\,s + 9.67 \times 10^6$; quadratic formula:

$$s = \frac{-7390 \pm \sqrt{(7390)^2 - 4(9.67 \times 10^6)}}{2} = \frac{-7390 \pm 3990}{2} \Rightarrow s = -1700,\ -5690$$

Factored form (leading coefficient in front):

$$H(s) = 9.67 \times 10^6 \, \frac{1}{(s + 1700)(s + 5690)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 271}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 905}}, \qquad K = 1$$

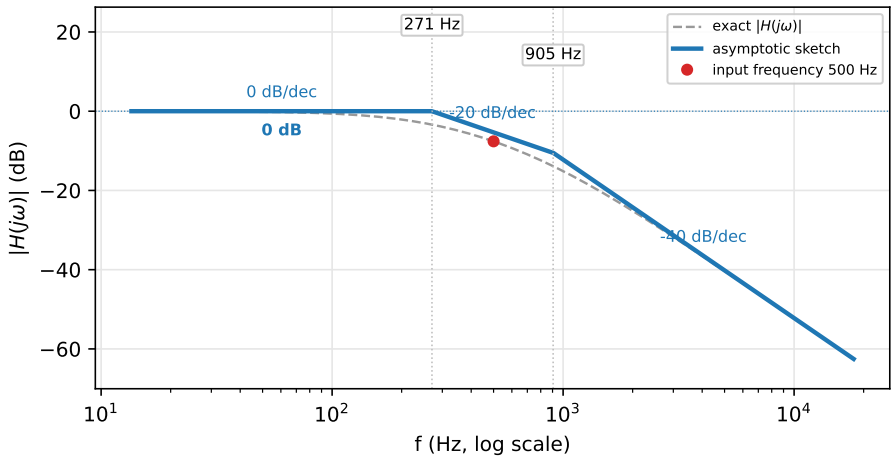
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1700	271 Hz	−20 dB/dec
pole	5690	905 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 271$ Hz	0 dB/dec (flat)	0 dB at end
271 Hz – 905 Hz	−20 dB/dec	0 dB at start, -10.5 dB at end
$f > 905$ Hz	−40 dB/dec	-10.5 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin\left(2\pi\cdot 500\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 271}}$	0.4759	-1.07
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 905}}$	0.8754	-0.505
total	**0.4166** (product)	**−1.58** (sum)

$$|H(j\omega_{in})| = 0.417, \qquad \angle H(j\omega_{in}) = -1.58\ \text{rad}$$
$$v_{o,ac}(t) = 0.417\times 0.5\,\sin\left(2\pi\cdot 500\,t - 0.794\right) = 0.208\,\sin\left(2\pi\cdot 500\,t - 0.794\right)$$

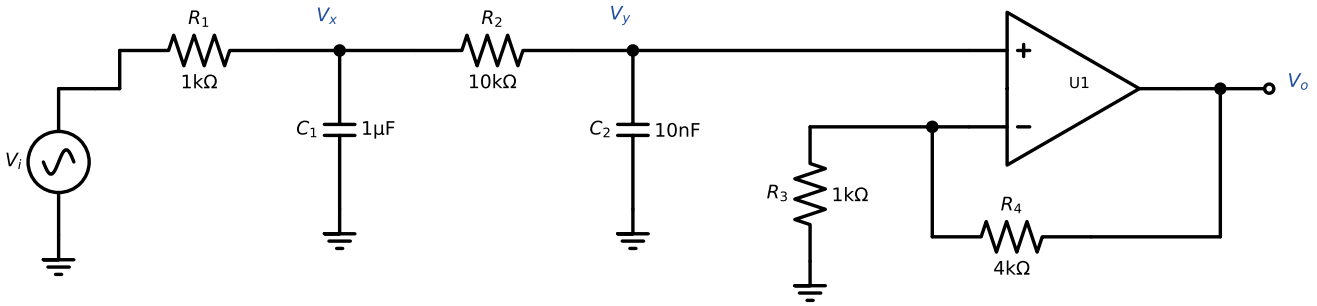
(phase: $\theta + \angle H = \frac{\pi}{4} + (-1.58) = -0.794\,\text{rad}$.)

$$\boxed{V_o(t) = 0.208\,\sin\left(2\pi\cdot 500\,t - 0.794\right)\ \text{V}}$$

Answer

$$H(s) = 1\,\frac{1}{1+\dfrac{s}{2\pi\cdot 271}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 905}} \quad \text{(corners: 271 Hz, 905 Hz)}$$
$$V_o(t) = 0.208\,\sin\left(2\pi\cdot 500\,t - 0.794\right)\ \text{V}$$

Solution 2B.12 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). A two-section passive ladder drives the + input. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= V_y): \quad \frac{V_y}{R_3} + \frac{V_y - V_o}{R_4} = 0$$

$$\text{(2) at } V_x: \quad \frac{V_x - V_i}{R_1} + sC_1V_x + \frac{V_x - V_y}{R_2} = 0$$

$$\text{(3) at } V_y: \quad \frac{V_y - V_x}{R_2} + sC_2V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_y = 0.2\,V_o$$

$$\text{from (2):} \quad V_x = \left(\frac{20}{s+1100}\right)V_o + \left(\frac{1000}{s+1100}\right)V_i$$

Substituting everything into (3) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{5\times10^7}{s^2 + 11100\,s + 10^7}$$

So:

$$H(s) = \frac{5\times10^7}{s^2 + 11100\,s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 11100\,s + 10^7$; quadratic formula:

$$s = \frac{-11100 \pm \sqrt{(11100)^2 - 4(10^7)}}{2} = \frac{-11100 \pm 9120}{2} \Rightarrow s = -989,\ -10100$$

Factored form (leading coefficient in front):

$$H(s) = 5 \times 10^7 \, \frac{1}{(s+989)(s+10100)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 157}} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.61\text{k}}}, \qquad K = 5$$

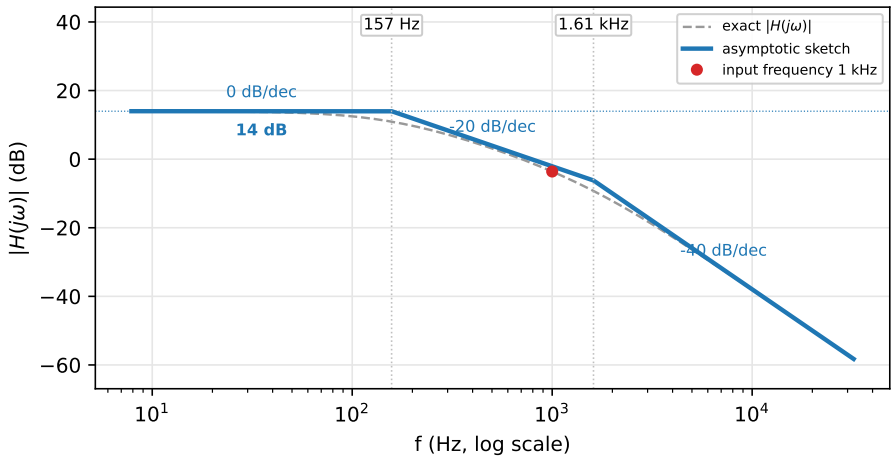
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	989	157 Hz	−20 dB/dec
pole	10100	1.61 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 157$ Hz	0 dB/dec (flat)	14 dB at end
157 Hz – 1.61 kHz	−20 dB/dec	14 dB at start, -6.21 dB at end
$f > 1.61$ kHz	−40 dB/dec	-6.21 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 157}}$	0.1555	-1.41
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.61\text{k}}}$	0.8494	-0.556
total	**0.6604** (product)	$-$**1.97** (sum)

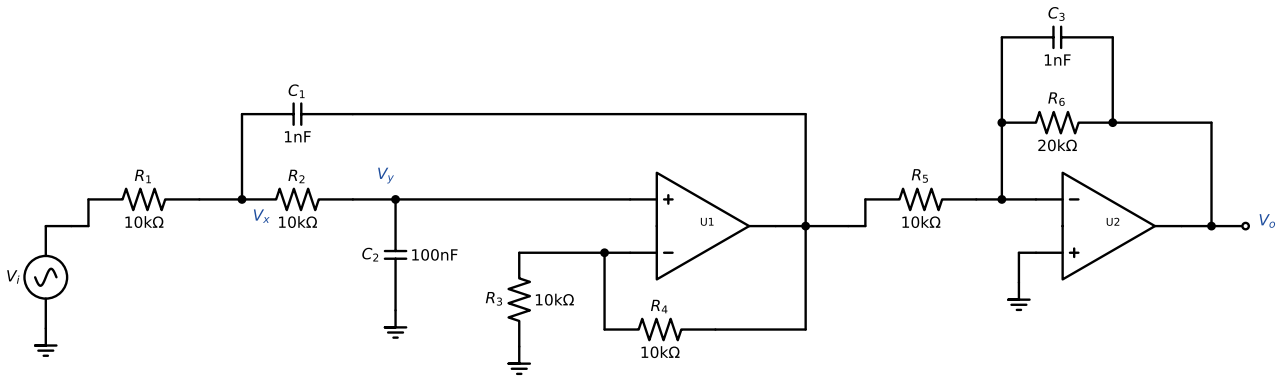
$$|H(j\omega_{in})| = 0.66, \qquad \angle H(j\omega_{in}) = -1.97 \text{ rad}$$
$$v_{o,ac}(t) = 0.66 \times 0.1 \sin(2\pi \cdot 1000\,t - 1.97) = 0.066 \sin(2\pi \cdot 1000\,t - 1.97)$$

$$\boxed{V_o(t) = 0.066 \sin(2\pi \cdot 1000\,t - 1.97) \text{ V}}$$

Answer

$$H(s) = 5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 157}}\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.61\text{k}}} \quad \text{(corners: 157 Hz, 1.61 kHz)}$$
$$V_o(t) = 0.066 \sin(2\pi \cdot 1000\,t - 1.97) \text{ V}$$

Solution 2B.13 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). Block 1 is not a standard block (nodal analysis, tools T3 and T4); the other blocks are standard (block formulas, tool T2). The blocks still multiply because every op-amp output is an ideal source.

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= V_y): \quad \frac{V_y}{R_3} + \frac{V_y - V_{o1}}{R_4} = 0$$

$$\text{(2) at } V_x: \quad \frac{V_x - V_i}{R_1} + \frac{V_x - V_y}{R_2} + sC_1 (V_x - V_{o1}) = 0$$

$$\text{(3) at } V_y: \quad \frac{V_y - V_x}{R_2} + sC_2 V_y = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_y = 0.5\, V_{o1}$$

$$\text{from (2):} \quad V_x = \left(\frac{s + 50000}{s + 2 \times 10^5}\right) V_{o1} + \left(\frac{10^5}{s + 2 \times 10^5}\right) V_i$$

Substituting everything into (3) leaves only V_{o1} and V_i.

$$\frac{V_{o1}}{V_i} = \frac{2 \times 10^8}{s^2 + 1.99 \times 10^5\, s + 10^8}$$

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{R_5}$ through R_5 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_5}$.

$$Z_{f2} = R_6 \parallel \frac{1}{sC_3} = \frac{10^9}{s + 50000}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_5} = \frac{-10^5}{s + 50000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-2 \times 10^{13}}{s^3 + 2.49 \times 10^5\, s^2 + 10^{10}\, s + 5 \times 10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^3 + 2.49 \times 10^5\, s^2 + 10^{10}\, s + 5 \times 10^{12}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -504,\ -50000,\ -1.98 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -2 \times 10^{13}\, \frac{1}{(s + 504)(s + 50000)(s + 1.98 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -4\, \frac{1}{1 + \frac{s}{2\pi \cdot 80.2}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 7.96\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 31.6\text{k}}}, \qquad K = -4$$

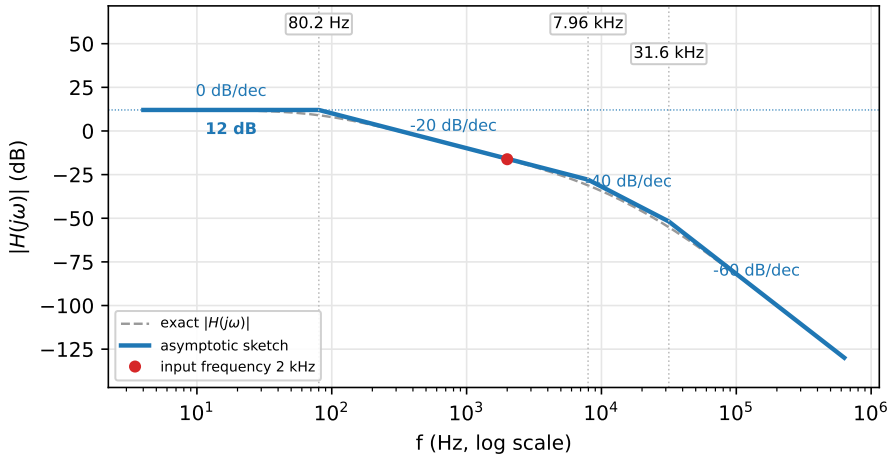
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	504	80.2 Hz	−20 dB/dec
pole	50000	7.96 kHz	−20 dB/dec
pole	1.98×10^5	31.6 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 4 = 12\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 80.2$ Hz	0 dB/dec (flat)	12 dB at end
80.2 Hz – 7.96 kHz	-20 dB/dec	12 dB at start, -27.9 dB at end
7.96 kHz – 31.6 kHz	-40 dB/dec	-27.9 dB at start, -51.8 dB at end
$f > 31.6$ kHz	-60 dB/dec	-51.8 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 4 = 12\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-4	4	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 80.2}}$	0.04006	-1.53
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 7.96\text{k}}}$	0.9698	-0.246
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 31.6\text{k}}}$	0.998	-0.0632
total	**0.1551** (product)	**-4.98** (sum)

$$|H(j\omega_{in})| = 0.155, \qquad \angle H(j\omega_{in}) = -4.98 \text{ rad}$$

$$v_{o,ac}(t) = 0.155 \times 0.1 \sin\left(2\pi \cdot 2000\, t - 4.98\right) = 0.0155 \sin\left(2\pi \cdot 2000\, t - 4.98\right)$$

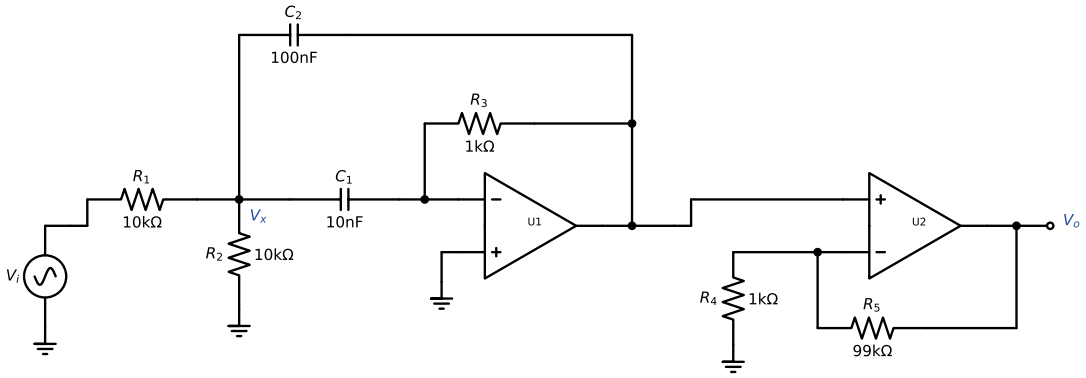
$$\boxed{V_o(t) = 0.0155 \sin\left(2\pi \cdot 2000\, t - 4.98\right) \text{ V}}$$

Answer

$$H(s) = -4 \frac{1}{1 + \dfrac{s}{2\pi \cdot 80.2}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 7.96\text{k}}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 31.6\text{k}}} \quad \text{(corners: 80.2 Hz, 7.96 kHz, 31.6 kHz)}$$

$$V_o(t) = 0.0155 \sin\left(2\pi \cdot 2000\, t - 4.98\right) \text{ V}$$

Solution 2B.14 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). Block 1 is not a standard block (nodal analysis, tools T3 and T4); the other blocks are standard (block formulas, tool T2). The blocks still multiply because every op-amp output is an ideal source.

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Multiple-feedback stage: node V_x is fed back from the output through a second branch. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (=0): \qquad sC_1 (0 - V_x) + \frac{0 - V_{o1}}{R_3} = 0$$

$$\text{(2) at } V_x: \qquad \frac{V_x - V_i}{R_1} + sC_1 V_x + sC_2 (V_x - V_{o1}) + \frac{V_x}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{-10^5}{s}\right) V_{o1}$$

Substituting everything into (2) leaves only V_{o1} and V_i.

$$\frac{V_{o1}}{V_i} = \frac{-1000\, s}{s^2 + 1.1 \times 10^5\, s + 2 \times 10^8}$$

Block 2 (U2). Non-inverting block: The + input is the block input: $V_+ = V_{o1}$. The op-amp forces $V_- = V_+$, and R_4, R_5 divide the output: $V_- = \frac{R_4}{R_4+R_5} V_o$, hence $V_o = \left(1 + \frac{R_5}{R_4}\right) V_+$.

$$\frac{V_o}{V_{o1}} = \left(1 + \frac{R_5}{R_4}\right) = 100$$

Multiply the blocks and simplify:

$$H(s) = \frac{-10^5\, s}{s^2 + 1.1 \times 10^5\, s + 2 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1.1 \times 10^5\, s + 2 \times 10^8$; quadratic formula:

$$s = \frac{-1.1 \times 10^5 \pm \sqrt{(1.1 \times 10^5)^2 - 4(2 \times 10^8)}}{2} = \frac{-1.1 \times 10^5 \pm 1.06 \times 10^5}{2} \Rightarrow s = -1850,\ -1.08 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = -10^5\, \frac{s}{(s + 1850)(s + 1.08 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -0.925\, \frac{s}{s + 2\pi \cdot 294}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 17.2\text{k}}}, \qquad K = -0.925$$

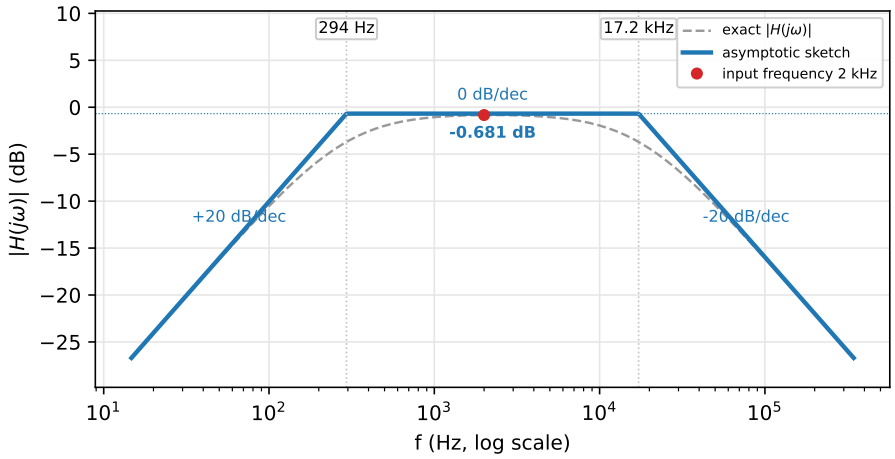
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1850	294 Hz	−20 dB/dec
pole	1.08×10^5	17.2 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 5 \times 10^{-4}\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 294$ Hz	+20 dB/dec	-0.681 dB at end
294 Hz – 17.2 kHz	0 dB/dec (flat)	-0.681 dB at start, -0.681 dB at end
$f > 17.2$ kHz	−20 dB/dec	-0.681 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}0.925 = -0.681\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.925	0.9246	-3.14
$\dfrac{s}{s+2\pi\cdot 294}$	0.9893	0.146
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 17.2\text{k}}}$	0.9933	-0.116
total	**0.9087** (product)	$-$**3.11** (sum)

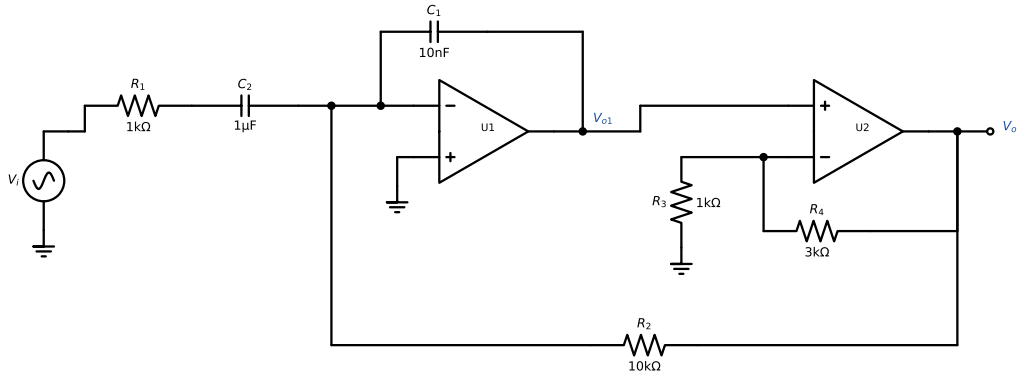
$$|H(j\omega_{in})| = 0.909, \qquad \angle H(j\omega_{in}) = -3.11\ \text{rad}$$
$$v_{o,ac}(t) = 0.909\times 0.05\sin(2\pi\cdot 2000\,t - 3.11) = 0.0454\sin(2\pi\cdot 2000\,t - 3.11)$$

$$\boxed{V_o(t) = 0.0454\sin(2\pi\cdot 2000\,t - 3.11)\ \text{V}}$$

Answer

$$H(s) = -0.925\,\frac{s}{s+2\pi\cdot 294}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 17.2\text{k}}} \quad \text{(corners: 294 Hz, 17.2 kHz)}$$
$$V_o(t) = 0.0454\sin(2\pi\cdot 2000\,t - 3.11)\ \text{V}$$

Solution 2B.15 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Two op-amps in a loop: U_2 amplifies V_{o1} and feeds it back into the summing node of U_1, so the stages cannot simply be multiplied. This is not a standard block, so use nodal analysis. Short names for the composite branches:

$$Z_a = R_1 + \frac{1}{sC_2} = \frac{1000\,s + 10^6}{s}$$

Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). U2: $V_- = V_+$, so V_n is the same voltage as V_{o1}. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_-\,(=0): \qquad \frac{0 - V_i}{Z_a} + sC_1\,(0 - V_{o1}) + \frac{0 - V_o}{R_2} = 0$$

$$\text{(2) at } V_n\,(=V_{o1}): \qquad \frac{V_{o1}}{R_3} + \frac{V_{o1} - V_o}{R_4} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_{o1} = \left(\frac{-10000}{s}\right) V_o + \left(\frac{-10^5}{s + 1000}\right) V_i$$

Substituting everything into (2) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-4 \times 10^5\,s}{s^2 + 41000\,s + 4 \times 10^7}$$

So:

$$H(s) = \frac{-4 \times 10^5\, s}{s^2 + 41000\, s + 4 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 41000\, s + 4 \times 10^7$; quadratic formula:

$$s = \frac{-41000 \pm \sqrt{(41000)^2 - 4(4 \times 10^7)}}{2} = \frac{-41000 \pm 39000}{2} \Rightarrow s = -1000,\ -40000$$

Factored form (leading coefficient in front):

$$H(s) = -4 \times 10^5\, \frac{s}{(s + 1000)(s + 40000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10\, \frac{s}{s + 2\pi \cdot 159}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 6.37\text{k}}}, \qquad K = -10$$

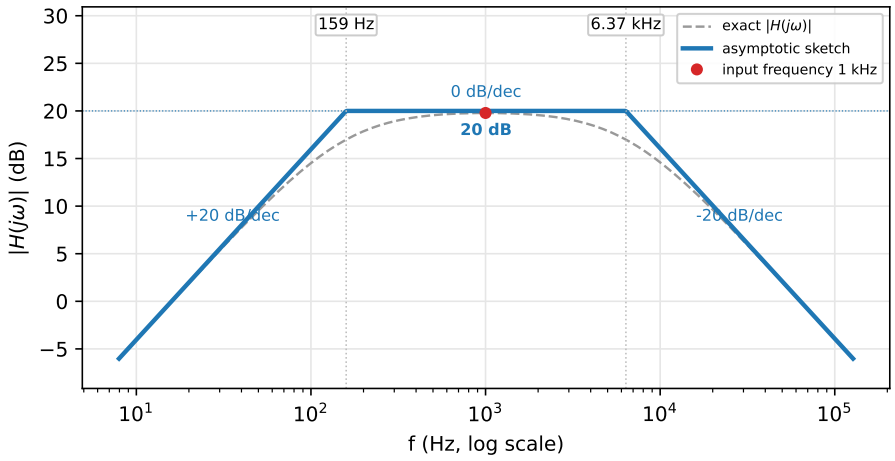
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	40000	6.37 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.01\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 159 Hz	+20 dB/dec	20 dB at end
159 Hz – 6.37 kHz	0 dB/dec (flat)	20 dB at start, 20 dB at end
$f >$ 6.37 kHz	−20 dB/dec	20 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 10 = 20$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	10	-3.14
$\dfrac{s}{s+2\pi\cdot 159}$	0.9876	0.158
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 6.37\text{k}}}$	0.9879	-0.156
total	**9.756** (product)	$\mathbf{-3.14}$ (sum)

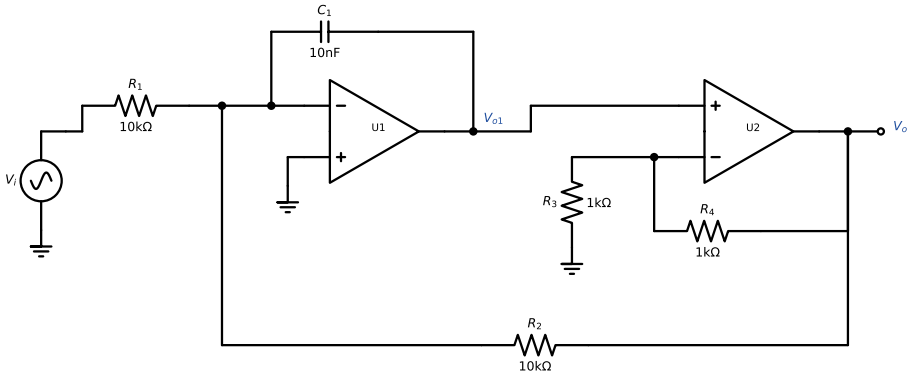
$$|H(j\omega_{in})| = 9.76, \qquad \angle H(j\omega_{in}) = -3.14 \text{ rad}$$
$$v_{o,ac}(t) = 9.76 \times 0.05\sin(2\pi\cdot 1000\,t - 3.14) = 0.488\sin(2\pi\cdot 1000\,t - 3.14)$$

$$\boxed{V_o(t) = 0.488\sin(2\pi\cdot 1000\,t - 3.14)\ \text{V}}$$

Answer

$$H(s) = -10\,\frac{s}{s+2\pi\cdot 159}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 6.37\text{k}}} \quad \text{(corners: 159 Hz, 6.37 kHz)}$$
$$V_o(t) = 0.488\sin(2\pi\cdot 1000\,t - 3.14)\ \text{V}$$

Solution 2B.16 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is not a standard block, so it is solved by nodal analysis (tools T3, T4).

Step 4 — Transfer function by nodal analysis

Block 1 (U1). Two op-amps in a loop: U_2 amplifies V_{o1} and feeds it back into the summing node of U_1, so the stages cannot simply be multiplied. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as 0 (virtual ground). U2: $V_- = V_+$, so V_n is the same voltage as V_{o1}. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= 0): \qquad \frac{0 - V_i}{R_1} + sC_1\,(0 - V_{o1}) + \frac{0 - V_o}{R_2} = 0$$

$$\text{(2) at } V_n\,(= V_{o1}): \qquad \frac{V_{o1}}{R_3} + \frac{V_{o1} - V_o}{R_4} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_{o1} = \left(\frac{-10000}{s}\right) V_o + \left(\frac{-10000}{s}\right) V_i$$

Substituting everything into (2) leaves only V_o and V_i.

$$\frac{V_o}{V_i} = \frac{-20000}{s + 20000}$$

So:

$$H(s) = \frac{-20000}{s + 20000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 20000$; $s + 20000 = 0 \Rightarrow s = -20000$.

Factored form (leading coefficient in front):

$$H(s) = -20000\,\frac{1}{(s + 20000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}}, \qquad K = -1$$

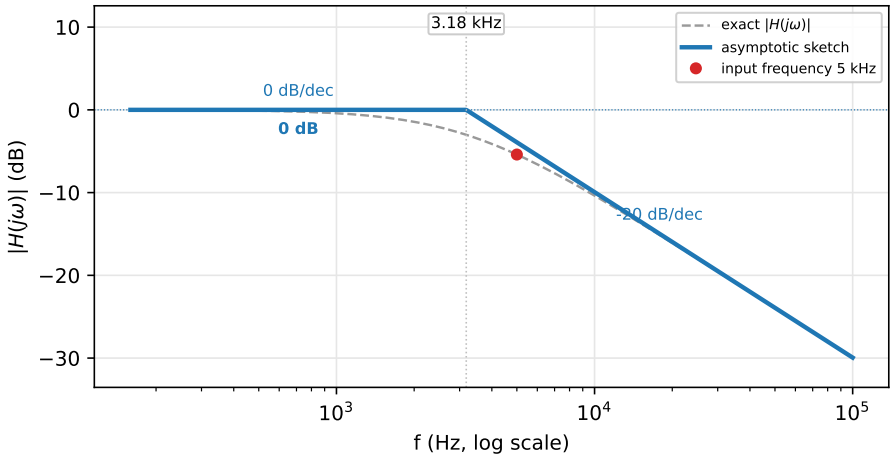
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	20000	3.18 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.18$ kHz	0 dB/dec (flat)	0 dB at end
$f > 3.18$ kHz	−20 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\cos\left(2\pi \cdot 5000\,t - \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1	1	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}}$	0.537	-1
total	**0.537** (product)	$-$**4.15** (sum)

$$|H(j\omega_{in})| = 0.537, \qquad \angle H(j\omega_{in}) = -4.15 \text{ rad}$$
$$v_{o,ac}(t) = 0.537 \times 0.2\,\cos\left(2\pi \cdot 5000\,t - 4.93\right) = 0.107\,\cos\left(2\pi \cdot 5000\,t - 4.93\right)$$

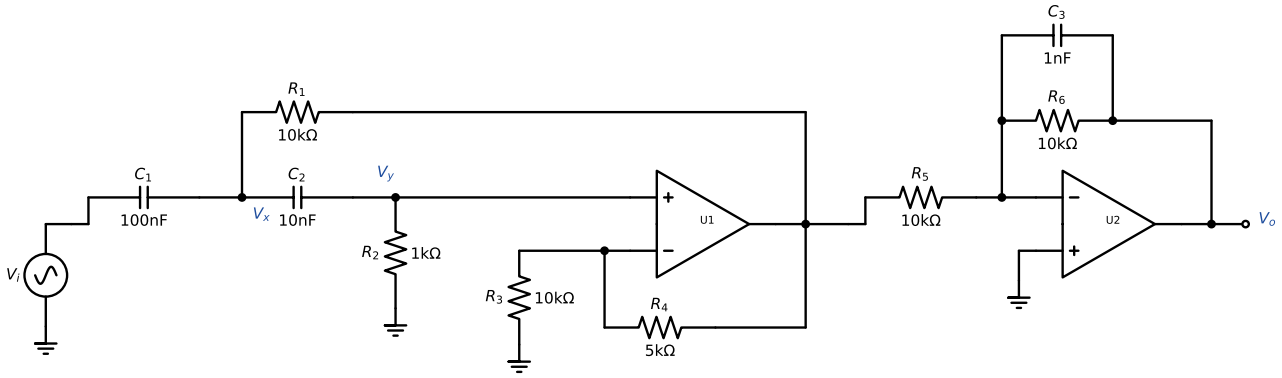
(phase: $\theta + \angle H = -\frac{\pi}{4} + (-4.15) = -4.93$ rad.)

$$\boxed{V_o(t) = 0.107\,\cos\left(2\pi \cdot 5000\,t - 4.93\right)\ \text{V}}$$

Answer

$$H(s) = -1\,\frac{1}{1+\dfrac{s}{2\pi \cdot 3.18\text{k}}} \quad \text{(corners: 3.18 kHz)}$$
$$V_o(t) = 0.107\,\cos\left(2\pi \cdot 5000\,t - 4.93\right)\ \text{V}$$

Solution 2B.17 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). Block 1 is not a standard block (nodal analysis, tools T3 and T4); the other blocks are standard (block formulas, tool T2). The blocks still multiply because every op-amp output is an ideal source.

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Sallen-Key stage: the network feeds back from the output to node V_x. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_- is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_- (= V_y): \qquad \frac{V_y}{R_3} + \frac{V_y - V_{o1}}{R_4} = 0$$

$$\text{(2) at } V_x: \qquad sC_1 (V_x - V_i) + sC_2 (V_x - V_y) + \frac{V_x - V_{o1}}{R_1} = 0$$

$$\text{(3) at } V_y: \qquad sC_2 (V_y - V_x) + \frac{V_y}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_y = 0.667\, V_{o1}$$

$$\text{from (2):} \quad V_x = \left(\frac{0.0606\, s + 909}{s + 909}\right) V_{o1} + \left(\frac{0.909\, s}{s + 909}\right) V_i$$

Substituting everything into (3) leaves only V_{o1} and V_i.

$$\frac{V_{o1}}{V_i} = \frac{1.5\,s^2}{s^2 + 1.09 \times 10^5\,s + 10^8}$$

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{R_5}$ through R_5 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_5}$.

$$Z_{f2} = R_6 \parallel \frac{1}{sC_3} = \frac{10^9}{s + 10^5}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_5} = \frac{-10^5}{s + 10^5}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-1.5 \times 10^5\,s^2}{s^3 + 2.1 \times 10^5\,s^2 + 1.1 \times 10^{10}\,s + 10^{13}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^3 + 2.1 \times 10^5\,s^2 + 1.1 \times 10^{10}\,s + 10^{13}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -921,\ -100000,\ -1.09 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = -1.5 \times 10^5\,\frac{s^2}{(s+921)(s+10^5)(s+1.09\times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1.38\,\frac{s}{s + 2\pi\cdot 147}\,\frac{s}{s + 2\pi\cdot 15.9\text{k}}\,\frac{1}{1 + \dfrac{s}{2\pi\cdot 17.3\text{k}}}, \qquad K = -1.38$$

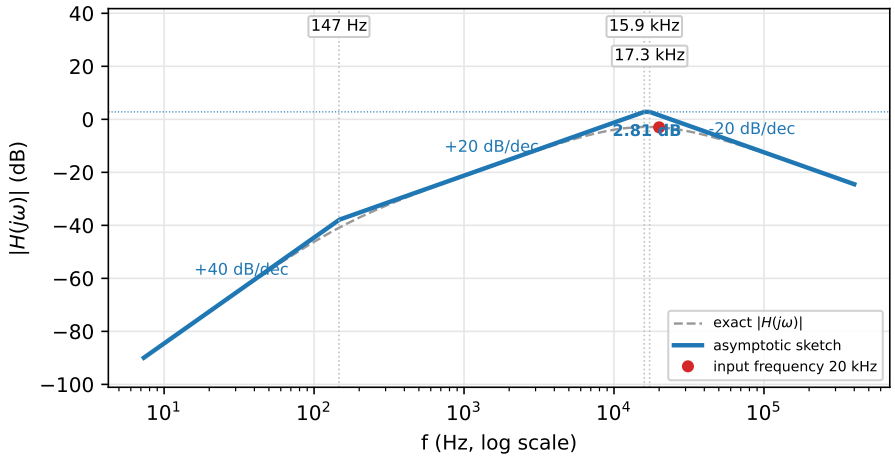
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	921	147 Hz	−20 dB/dec
pole	10^5	15.9 kHz	−20 dB/dec
pole	1.09×10^5	17.3 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.5 \times 10^{-8}\,s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 147$ Hz	+40 dB/dec	-37.9 dB at end
147 Hz – 15.9 kHz	+20 dB/dec	-37.9 dB at start, 2.81 dB at end
15.9 kHz – 17.3 kHz	0 dB/dec (flat)	2.81 dB at start, 2.81 dB at end
$f > 17.3$ kHz	−20 dB/dec	2.81 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1.38 = 2.81$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 20000\,t)$, so $\omega_{in} = 2\pi\cdot 20000 = 1.26\times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.38	1.381	-3.14
$\dfrac{s}{s+2\pi\cdot 147}$	1	0.00733
$\dfrac{s}{s+2\pi\cdot 15.9\text{k}}$	0.7825	0.672
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 17.3\text{k}}}$	0.6538	-0.858
total	**0.7067** (product)	$\mathbf{-3.32}$ (sum)

$$|H(j\omega_{in})| = 0.707, \qquad \angle H(j\omega_{in}) = -3.32 \text{ rad}$$
$$v_{o,ac}(t) = 0.707 \times 0.1 \sin(2\pi\cdot 20000\,t - 3.32) = 0.0707 \sin(2\pi\cdot 20000\,t - 3.32)$$

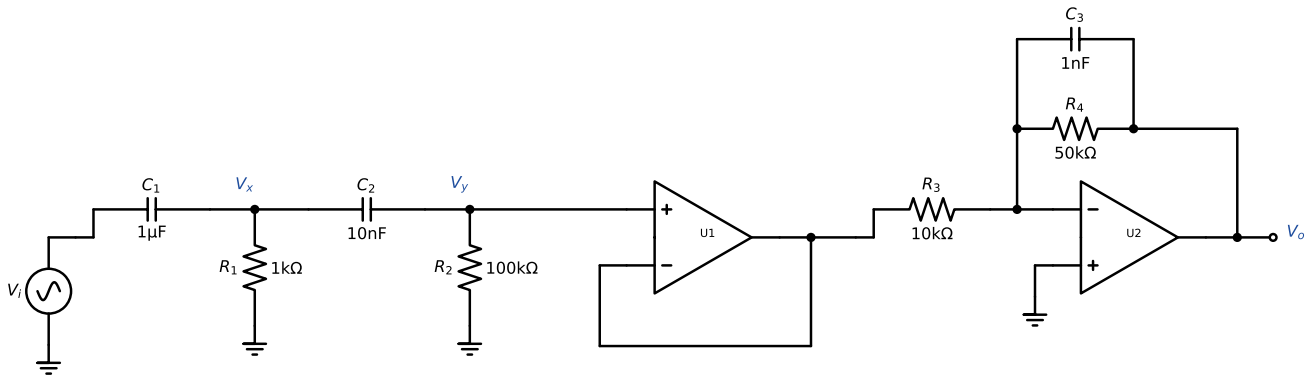
$$\boxed{V_o(t) = 0.0707 \sin\left(2\pi \cdot 20000\, t - 3.32\right) \text{ V}}$$

Answer

$$H(s) = -1.38 \frac{s}{s + 2\pi \cdot 147} \frac{s}{s + 2\pi \cdot 15.9\text{k}} \frac{1}{1 + \dfrac{s}{2\pi \cdot 17.3\text{k}}}$$ (corners: 147 Hz, 15.9 kHz, 17.3 kHz)

$$V_o(t) = 0.0707 \sin\left(2\pi \cdot 20000\, t - 3.32\right) \text{ V}$$

Solution 2B.18 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). Block 1 is not a standard block (nodal analysis, tools T3 and T4); the other blocks are standard (block formulas, tool T2). The blocks still multiply because every op-amp output is an ideal source.

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). A two-section passive ladder drives the $+$ input. This is not a standard block, so use nodal analysis. Virtual short (tool T1): U1: $V_- = V_+$, so V_{o1} is the same voltage as V_y. KCL (tool T3) at every node that is not an op-amp output or a source; no current flows into the op-amp inputs, so the op-amps do not appear in these sums:

$$\text{(1) at } V_x: \qquad sC_1\,(V_x - V_i) + \frac{V_x}{R_1} + sC_2\,(V_x - V_y) = 0$$

$$\text{(2) at } V_y: \qquad sC_2\,(V_y - V_x) + \frac{V_y}{R_2} = 0$$

Solve them one at a time (tool T4): take the equation with the fewest unknowns, solve it for one node voltage and substitute that into the remaining equations. With the component values:

$$\text{from (1):} \quad V_x = \left(\frac{0.0099\,s}{s + 990}\right) V_y + \left(\frac{0.99\,s}{s + 990}\right) V_i$$

Substituting everything into (2) leaves only V_y and V_i. Since $V_{o1} = V_y$ (virtual short),

$$\frac{V_{o1}}{V_i} = \frac{V_y}{V_i} = \frac{s^2}{s^2 + 2010\,s + 10^6}$$

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{R_3}$ through R_3 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_3}$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_3} = \frac{10^9}{s + 20000}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_3} = \frac{-10^5}{s + 20000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-10^5\, s^2}{s^3 + 22000\, s^2 + 4.12 \times 10^7\, s + 2 \times 10^{10}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^3 + 22000\, s^2 + 4.12 \times 10^7\, s + 2 \times 10^{10}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -905,\ -1110,\ -20000$.

Factored form (leading coefficient in front):

$$H(s) = -10^5 \frac{s^2}{(s + 905)(s + 1110)(s + 20000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -5 \frac{s}{s + 2\pi \cdot 144} \frac{s}{s + 2\pi \cdot 176} \frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}}, \qquad K = -5$$

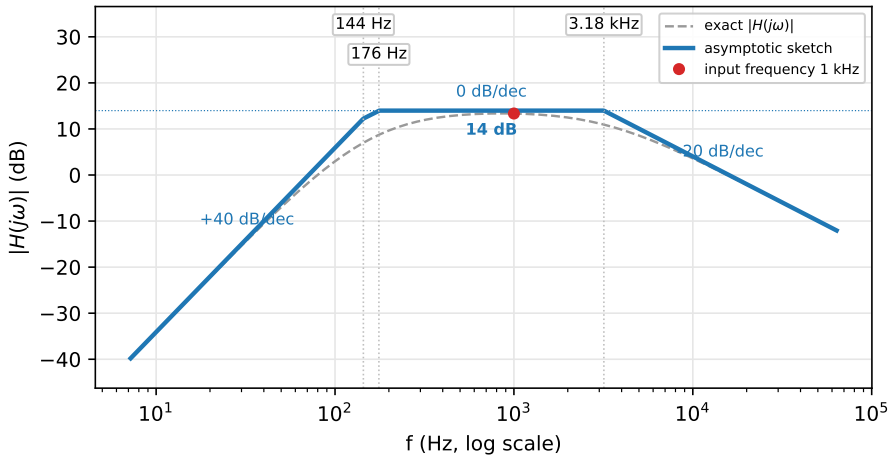
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	905	144 Hz	−20 dB/dec
pole	1110	176 Hz	−20 dB/dec
pole	20000	3.18 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 5 \times 10^{-6}\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 144 Hz	+40 dB/dec	12.2 dB at end
144 Hz – 176 Hz	+20 dB/dec	12.2 dB at start, 14 dB at end
176 Hz – 3.18 kHz	0 dB/dec (flat)	14 dB at start, 14 dB at end
$f >$ 3.18 kHz	−20 dB/dec	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin\left(2\pi\cdot 1000\,t + \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{s}{s+2\pi\cdot 144}$	0.9898	0.143
$\dfrac{s}{s+2\pi\cdot 176}$	0.9849	0.174
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.18\text{k}}}$	0.954	-0.304
total	**4.65** (product)	$\mathbf{-3.13}$ (sum)

$$|H(j\omega_{in})| = 4.65, \qquad \angle H(j\omega_{in}) = -3.13\ \text{rad}$$
$$v_{o,ac}(t) = 4.65\times 0.05\,\sin\left(2\pi\cdot 1000\,t - 2.08\right) = 0.233\,\sin\left(2\pi\cdot 1000\,t - 2.08\right)$$

(phase: $\theta + \angle H = \frac{\pi}{3} + (-3.13) = -2.08\,\text{rad}$.)

$$\boxed{V_o(t) = 0.233\,\sin\left(2\pi\cdot 1000\,t - 2.08\right)\ \text{V}}$$

Answer

$$H(s) = -5\,\frac{s}{s + 2\pi \cdot 144}\,\frac{s}{s + 2\pi \cdot 176}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}} \quad \text{(corners: 144 Hz, 176 Hz, 3.18 kHz)}$$

$$V_o(t) = 0.233\,\sin\left(2\pi \cdot 1000\,t - 2.08\right)\ \text{V}$$